

Sharp Targeting with Generated States

Boundary Traces and the Validation Frontier

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Abstract

Many policy rules rank units using generated scores. For sharp targeting, ordinary prediction accuracy is the wrong validation criterion: errors far from the cutoff have little first-order welfare relevance. This paper shows that the relevant validation object is boundary-local. Fixed cutoffs depend on a welfare-weighted boundary trace, fixed-capacity rules on a boundary covariance, and learned cutoffs on a residualized trace-slope. For learned cutoffs, evaluation-sample cube-root uncertainty and root- m validation uncertainty meet at the frontier $m \asymp n^{2/3}$. When the residualized trace-slope is nonzero, validation error is first order at this frontier and negligible only when $m \gg n^{2/3}$. For fixed representations followed by a finite-dimensional calibration layer, the paper gives primitive influence functions and feasible validation-adjusted inference.

KEYWORDS: Policy learning, generated regressors, algorithmic targeting, boundary trace, validation frontier, calibrated generated states.

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1 Introduction

Many policies rank people, firms, or places using generated states: risk scores, calibrated forecasts, fixed text indices, and factor estimates. Let Z denote the raw covariates or records used to build a score. The implemented rule acts on the generated state $\widehat{X} = \widehat{g}(Z)$. Welfare is indexed by a benchmark state $X = g_0(Z)$ and a welfare score Γ . When the rule is sharp, the first-stage error

$$\Delta = \widehat{X} - X$$

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matters only through units near the policy boundary. Global prediction loss therefore does not target the margin that matters for welfare. The relevant object is local and welfare weighted. The primitive inference theory below is restricted to a finite-dimensional calibration layer applied to a fixed upstream representation. End-to-end retraining enters only when that upstream step has a boundary-local influence expansion.

The main result is a validation frontier for learned sharp cutoffs. Let $\mathbb{G}_\partial(u)$ denote the usual two-sided cube-root threshold process from the evaluation sample. Let N_g denote the root- m validation noise in the residualized trace-slope. The local experiment for the latent-equivalent learned cutoff is

$$\arg \max_{u \in \mathbb{R}} \left\{ \mathbb{G}_\partial(u) - \frac{1}{2} \kappa u^2 + u \left(\beta_b + \sqrt{\lambda_g} \sigma_{T,\perp} N_g \right) \right\}, \quad \lambda_g = \lim n^{2/3}/m.$$

Here u is the $n^{1/3}$ -scaled local cutoff coordinate. The term $\mathbb{G}_\partial(u)$ is the evaluation-sample cube-root noise. The term N_g is the validation-sample trace-slope noise. The coefficient $\sqrt{\lambda_g}$ is the scale at which the two sources meet. The curvature $\kappa > 0$ is the local welfare curvature, β_b is a possible deterministic first-stage trace-slope bias, and $\sigma_{T,\perp}$ is the standard deviation of the residualized trace-slope influence function. If the residualized trace-slope is zero, this validation term disappears. Otherwise root- m validation error is first order at $m \asymp n^{2/3}$ and is negligible only when $m \gg n^{2/3}$.

A simple example illustrates why the trace-slope, not global RMSE, is the relevant object. Let $X \sim U[-1, 1]$, let $G \in \{-1, 1\}$ be independent of X , and let $\Gamma = X(1 + \lambda G) + \varepsilon$, where ε is mean-zero noise independent of (X, G) . With f_X denoting the density of X , at the optimal cutoff $c_0 = 0$ the fixed-cutoff trace in the direction G is zero:

$$f_X(0) E[\Gamma G \mid X = 0] = 0.$$

Writing $\mathcal{T}_0(\cdot)$ for the boundary trace-slope at $c_0 = 0$, this derivative is nonzero:

$$\mathcal{T}_0(G) = \lambda/2.$$

A common location shift of the score can be absorbed by the learned numerical cutoff. The heterogeneous shift $X + \theta G$ cannot. If $\sqrt{m} \hat{\theta}$ is asymptotically normal and $n^{2/3}/m \rightarrow \lambda_g$, then the drifted local argmax has the same marginal law as the oracle argmax plus an independent Gaussian validation component. Section 4.1 gives the full calculation.

The paper has two main contributions. First, it identifies the boundary objects that govern

generated-state targeting. A fixed cutoff depends on the boundary trace

$$f_X(c)E[\Gamma(\widehat{X} - X) \mid X = c].$$

This trace is continuous in a boundary seminorm but not in global $L^2(P_Z)$. For fixed-capacity targeting at selected share q , write c_q for the latent cutoff that selects that share. The cutoff adjusts to keep the selected share fixed, and the first-order object becomes

$$f_X(c_q)\text{Cov}(\Gamma, \Delta \mid X = c_q),$$

so a common local level shift is removed and the boundary covariance remains.

Second, the paper derives the learned-cutoff trace-slope experiment. If the first-stage trace-slope error is $O_p(r_m)$, it is negligible only when $n^{1/3}r_m \rightarrow 0$. Root- m validation gives the frontier $m \gg n^{2/3}$. The paper gives the local limit experiment, a rate-sharp lower bound, an additive cutoff-variance representation, and primitive influence functions for a finite-dimensional calibration layer.

The primitive theory for feasible inference applies to fixed or frozen representations followed by a finite-dimensional calibration layer, such as a published risk score, a fixed factor score, or a frozen text embedding (Gentzkow et al., 2019; Ash and Hansen, 2023; Ludwig et al., 2026; Battaglia et al., 2024). It does not cover end-to-end retraining of a large representation. If the upstream representation is retrained, its uncertainty must enter the same influence-function expansion; a modular extension can do this when such an expansion is available.

The supplemental appendix reports two diagnostics. A semi-synthetic exercise uses the Alatas panel only to fix the boundary shape and then injects artificial generated-state errors. At the same global RMSE, the boundary-local perturbation has about 6.5 times the residualized trace-slope and 2.8 times the frontier cutoff variance. A separate poverty-targeting diagnostic shows how the fixed-capacity boundary covariance is computed and records scope limitations. Neither exercise is used as empirical validation.

The closest literatures differ in the object they validate. Threshold-policy work studies regret when the running variable is observed (Crippa, 2025). Statistical treatment-rule and policy-learning work studies how to choose welfare-improving rules from data (Manski, 2004; Hirano and Porter, 2009; Kitagawa and Tetenov, 2018; Athey and Wager, 2021; Mbakop and Tabord-Meehan, 2021). Generated-regressor and two-step papers study first-stage error in smooth or nonsmooth second-stage estimators (Pagan, 1984; Murphy and Topel, 2002;

Newey, 1994; Newey and McFadden, 1994; Hahn and Ridder, 2013; Sun, 2026). Cube-root and local empirical-process tools underlie the learned-cutoff limit (Kim and Pollard, 1990; Andrews, 1992; Cattaneo et al., 2020). Measurement-error RD studies causal discontinuities with a mismeasured assignment variable (Davezies and Le Barbanchon, 2017; Pei and Shen, 2017; Bartalotti et al., 2021). Orthogonal-score and double-machine-learning work clarifies when nuisance estimates can be debiased in regular problems (Chernozhukov et al., 2018; Argañaraz and Escanciano, 2025). These literatures provide the benchmarks, but the object here is different: normative assignment welfare when the policy boundary itself is built from a generated state.

The closest recent comparison is Opocher (2026); related efficiency work studies regret distributions for learned randomized policies (Fang et al., 2025). Both papers ask how information about policy-relevant states affects learned rules. Opocher’s accuracy requirement is global: the unconditional RMSE of the latent-trait estimate enters his regret bound as a term that is additively separate from a policy-class complexity term, with both terms traded off at the data-collection design stage. His margin condition is also a boundary condition, but it enters through a uniform margin constant that scales global RMSE. The object here is local to the sharp policy boundary: only welfare-weighted error near the cutoff is first order. Section 4 gives a same-RMSE contrast. Two generated scores can have the same global prediction loss, but only the one with nonzero welfare-weighted trace-slope changes the first-order learned-cutoff experiment.

The closest technical comparison is Sun (2026). That paper studies cube-root inference when the boundary regressor is a generated high-frequency latent state. Sun’s drift-robustness result allows local, common-scale first-stage error under payoff and boundary conditional-centering conditions. The robustness comes from cube-root locality and the oracle first-order condition, not from the error being a common location shift that the cutoff absorbs. In the notation below, the drift-robust face is $\mathcal{T}_{c_0}^\perp = 0$. Zero fixed-cutoff trace at a single boundary point is not enough; the residualized trace-slope must vanish. Sun-type payoff-orthogonal errors reach that face when the welfare-weighted trace is locally flat at the boundary, and genuine common-location shifts reach it through cutoff reparameterization. This paper studies the complementary heterogeneous directions, where $\mathcal{T}_{c_0}^\perp \neq 0$ and $n^{2/3}/m \rightarrow \lambda_g > 0$. Then validation noise enters the learned-cutoff local experiment. Measurement-error RD papers estimate causal discontinuities under a noisy assignment variable. Here the estimand is normative assignment welfare, and the boundary is a policy choice rather than a fixed institutional discontinuity.

Section 2 defines the welfare objects and data regimes. Section 3 gives the fixed-cutoff and fixed-capacity theory. Section 4 derives the learned-cutoff and calibrated-score results. Section 5 concludes. The main appendix gives the core proofs. Additional proofs, semi-synthetic diagnostics, simulation tables, the poverty-targeting diagnostic, and implementation details are in the supplemental appendix.

2 Model and Welfare Objects

This section separates samples, states, and welfare objects. The evaluation sample is

$$\mathcal{E}_n = \{O_i\}_{i=1}^n,$$

which is used to evaluate welfare and learn the cutoff. The validation or calibration sample is

$$\mathcal{V}_m = \{S_j\}_{j=1}^m,$$

which is used to construct or calibrate the generated state. The main product experiment is stated for an honest split, $\mathcal{E}_n \cap \mathcal{V}_m = \emptyset$, so that the evaluation empirical process and the validation influence function are asymptotically independent. K-fold cross-fitting is allowed only when each fold’s evaluation criterion uses first-stage and nuisance objects estimated on observations excluded from that fold; otherwise an additional covariance term can enter the local experiment.

The data regime determines what the validation sample identifies. If a benchmark state is observed near the boundary, the validation sample identifies the boundary trace and the fixed-capacity boundary covariance. If the generated state is a finite-dimensional calibration layer, calibration moments identify the influence functions for the trace and trace-slope. If the upstream representation is fixed, such as a fixed text index, factor score, risk score, or embedding, the feasible calibration-layer theory applies. If the upstream representation is retrained, the theory applies only when the upstream influence function is included in the calibration expansion.

Throughout, X is the benchmark state, $\widehat{X}$ is the implemented generated state, $\Delta = \widehat{X} - X$ is the generated-state error, and Γ is the welfare score. The sign convention for Δ is fixed in all theoretical and empirical statements.

Observability conventions. The oracle welfare $M(c) = E[\Gamma \mathbf{1}\{X \geq c\}]$ is the benchmark object. In benchmark-state validation regimes, X is observed in the validation fold near the boundary. In calibration-layer regimes, $X = q_{\theta_0}(Z)$ is identified through calibration

moments. In treatment-effect score regimes, Γ is either an observed welfare label or a cross-fitted score built from variables in the evaluation fold. If constructing Γ itself uses a generated X , that error is part of the localized nuisance remainder in Lemma 4.

Remark 1 (Sample splitting and the validation frontier). *With a total budget $N = n + m$, the leading evaluation regret for a learned sharp cutoff is $n^{-2/3}$, while root- m validation contributes m^{-1} through the trace-slope variance. Equalizing the two gives $m \asymp n^{2/3}$. Thus the validation share can vanish, $m/N \rightarrow 0$, but the validation sample must still grow at the $N^{2/3}$ scale to remove first-stage validation noise from the first-order cutoff law. If the same observations are used for validation and evaluation without cross-fitting, the threshold process $\mathbb{G}_\partial(u)$ and validation Gaussian N_g need not be independent. The frontier scale is still the natural one, but the limit experiment may contain an additional covariance term. The main statements use the honest split; cross-fitting is an extension that must preserve foldwise independence of the local criterion from the first-stage objects used inside that fold.*

The distinction from smooth generated-regressor work is central. Hahn–Ridder and Murphy–Topel corrections treat differentiable second-stage moments. Here the second stage contains $\mathbf{1}\{X \geq c\}$. Its derivative is a boundary distribution, not an L^2 Riesz representer. Standard smooth two-step variance formulas therefore do not remove generated-state error in sharp targeting.

2.1 Welfare objects and generated states

Let $O = (Y, D, Z, A)$ denote source-population data. $D \in \{0, 1\}$ is treatment, Y is outcome, A is an ordinary identification covariate, and Z is a raw high-dimensional object. Potential outcomes satisfy

$$Y = DY(1) + (1 - D)Y(0).$$

The latent economic state is scalar in the baseline model,

$$X = g_0(Z) \in \mathbb{R},$$

and the generated state is

$$\widehat{X} = \widehat{g}(Z) = X + \Delta.$$

Let $\Gamma = \Gamma(O; \eta_0)$ be a welfare score such that

$$\mathbb{E}[\Gamma \mid A, X] = \mathbb{E}[Y(1) - Y(0) \mid A, X]$$

under the identifying assumptions. For example, under unconfoundedness and overlap, one may use the doubly robust score

$$\Gamma(O; \eta_0) = \mu_1(A, X) - \mu_0(A, X) + \frac{D\{Y - \mu_1(A, X)\}}{e(A, X)} - \frac{(1 - D)\{Y - \mu_0(A, X)\}}{1 - e(A, X)}.$$

The notation below suppresses the nuisance argument when no confusion arises.

For a policy $\pi : \mathbb{R} \rightarrow [0, 1]$, define two welfare objects.

Latent-state benchmark.

$$V^*(\pi) = \mathbb{E}[\Gamma\pi(X)].$$

This is the welfare of a rule that acts on the latent economic state.

Implemented generated-state welfare.

$$V^g(\pi) = \mathbb{E}[\Gamma\pi(\widehat{X})].$$

This is the welfare of the rule that is actually implemented.

The generated-state distortion is

$$B_g(\pi) = V^g(\pi) - V^*(\pi) = \mathbb{E}[\Gamma\{\pi(\widehat{X}) - \pi(X)\}]. \quad (1)$$

For a threshold $\pi_c(x) = \mathbf{1}\{x \geq c\}$, the paper studies

$$B_g(c) = \mathbb{E}[\Gamma\{\mathbf{1}(X + \Delta \geq c) - \mathbf{1}(X \geq c)\}]. \quad (2)$$

For learned cutoffs, a scalar cutoff can absorb a local common shift in the generated score.

For any perturbation v , define

$$a(v) = E[v(Z) \mid X = c_0], \quad v_v(Z) = v(Z) - a(v),$$

and define the residualized same-boundary distortion

$$B_{v_v}(c) = \mathbb{E}[\Gamma\{\mathbf{1}(X + v_v(Z) \geq c) - \mathbf{1}(X \geq c)\}]. \quad (3)$$

The raw object B_g is the same-numerical-cutoff distortion. The learned-cutoff experiment below uses B_{v_v} because v_v is the part of the generated-state perturbation that cannot be absorbed by moving the scalar cutoff.

3 Boundary Trace Theory

This section isolates the first nonstandard feature of the problem. Even before sampling error and nuisance estimation, a generated state creates a partial-identification and minimax problem near the policy boundary.

3.1 Sharp partial identification under calibration radius

Suppose the researcher observes $(\widehat{X}, \Gamma)$ and knows that the latent state is within radius r :

$$|X - \widehat{X}| \leq r \quad \text{almost surely.}$$

For a fixed threshold c , define $\Gamma_+ = \max\{\Gamma, 0\}$ and $\Gamma_- = \min\{\Gamma, 0\}$.

Theorem 1 (Sharp boundary-band identified set). *Under $|X - \widehat{X}| \leq r$, the sharp identified set for $V^*(c) = \mathbb{E}[\Gamma \mathbf{1}\{X \geq c\}]$ is $[L_r(c), U_r(c)]$, where*

$$L_r(c) = \mathbb{E}\left[\Gamma \mathbf{1}\{\widehat{X} \geq c + r\} + \Gamma_- \mathbf{1}\{c - r \leq \widehat{X} < c + r\}\right],$$

$$U_r(c) = \mathbb{E}\left[\Gamma \mathbf{1}\{\widehat{X} \geq c + r\} + \Gamma_+ \mathbf{1}\{c - r \leq \widehat{X} < c + r\}\right].$$

The width is

$$U_r(c) - L_r(c) = \mathbb{E}\left[|\Gamma| \mathbf{1}\{c - r \leq \widehat{X} < c + r\}\right].$$

The theorem says that latent-state welfare uncertainty is exactly boundary-band welfare mass. The result is not a generic measurement-error statement: the only units that matter for the policy benchmark are those whose generated state lies within the calibration radius of the cutoff.

3.2 Prediction-loss modulus

The previous theorem uses a uniform calibration radius. Many representation models are judged by L^p prediction loss. The next result shows why this is not enough for sharp targeting.

Let X have density f_X . Fix $c = 0$ for notational simplicity and focus on upward perturbations from below the boundary. Define

$$\omega_p(r) = \sup_u \mathbb{E}[\Gamma \{\mathbf{1}(X + u(X) \geq 0) - \mathbf{1}(X \geq 0)\}]$$

where the supremum is over measurable $u \geq 0$ satisfying $\mathbb{E}|u(X)|^p \leq r^p$.

Assumption 1 (Local boundary welfare density). *There exist constants $0 < \underline{K} \leq \bar{K} < \infty$, $0 < \underline{f} \leq \bar{f} < \infty$, $\bar{a} > 0$, and $0 \leq \alpha < p$ such that for $0 < t < \bar{a}$,*

$$\underline{K}t^\alpha \leq w(t) := f_X(-t)\mathbb{E}[\Gamma \mid X = -t] \leq \bar{K}t^\alpha, \quad \underline{f} \leq f_X(-t) \leq \bar{f}.$$

Assumption 2 (Remote dominance). *For the global modulus, the remote region is dominated: $\mathbb{E}[|\Gamma| \mid X = x] \leq \bar{G}$ for $|x| \geq \bar{a}$. Then any remote contribution is $O(r^p)$, because crossing the boundary from that region requires displacement at least $\bar{a}$. Since $0 \leq \alpha < p$, $r^p = o(r^{p(\alpha+1)/(p+1)})$. For the local modulus, admissible perturbations are restricted to $(-\bar{a}, 0)$ and this assumption is unnecessary.*

Lemma 1 (Boundary-tube rearrangement). *Let $A \subset (0, \bar{a})$ be measurable and suppose*

$$\int_A t^p f_X(-t) dt \leq R.$$

Under Assumption 1,

$$\int_A w(t) dt \leq CR^{(\alpha+1)/(p+1)}$$

for a constant C independent of A and small R . The rate is attained, up to constants, by the boundary interval $A = [0, a_R]$ with $a_R \asymp R^{1/(p+1)}$.

Theorem 2 (Prediction-loss modulus). *Under Assumptions 1 and 2, as $r \downarrow 0$,*

$$\omega_p(r) \asymp r^{p(\alpha+1)/(p+1)}.$$

In particular, if $p = 2$ and $\alpha = 0$, the largest one-sided welfare loss under RMSE r is of order $r^{2/3}$. The maintained range $\alpha < p$ is the boundary-dominated case. At $\alpha = p$ the boundary and remote Lipschitz contributions have the same order; for $\alpha > p$, global Lipschitz behavior can dominate the boundary tube.

This result separates prediction accuracy from policy accuracy. A representation model can have small L^2 error and still have welfare distortion much larger than the RMSE rate when its error is concentrated near the policy boundary. The theorem is a one-dimensional boundary-tube benchmark: the perturbation is $u(X)$. Arbitrary Z -dependent representation errors can exploit within- X heterogeneity in Γ and require conditional restrictions or the boundary trace formulas below.

Relation to plug-in classification. The modulus is close to margin-condition calculations in plug-in classification (Mammen and Tsybakov, 1999; Tsybakov, 2004; Audibert and

Tsybakov, 2007). If Γ were a nonnegative excess-loss weight and the policy boundary coincided with the Bayes decision boundary, the same tube calculation would reproduce the familiar conversion from first-stage norm error to excess classification risk. The present object differs in three ways. First, c may be a policy or capacity cutoff rather than the zero of the Bayes score; then $\mathbb{E}[\Gamma \mid X = c]$ need not vanish, and fixed-threshold welfare has the first-order trace $f_X(c)\mathbb{E}[\Gamma \Delta \mid X = c]$. Second, Γ is a signed welfare score, so local gains and losses may cancel at the boundary. Third, once the cutoff is learned, the relevant first-stage object is not the margin density alone but the trace-slope $\partial_c\{f_X(c)\mathbb{E}[\Gamma \Delta \mid X = c]\}_{c=c_0}$, which drives the $m \asymp n^{2/3}$ validation frontier below. Thus Theorem 2 should be read as the boundary-tube bridge between prediction loss and welfare loss; the new policy-learning object is the welfare trace and its slope.

3.3 Boundary traces and L^2 non-regularity

Define the functional

$$\Theta_c(g) = \mathbb{E}[\Gamma \mathbf{1}\{g(Z) \geq c\}], \quad X = g_0(Z).$$

If generated-state policy welfare were an ordinary smooth generated-regressor problem, the derivative of Θ_c with respect to g would be representable by an $L^2(P_Z)$ influence function. The next theorem shows that this is false.

Assumption 3 (Trace directions and L^2 topology). *X has a density continuous and positive at c . Let $\mathcal{D}$ be a linear class of directions such that, for each $h \in \mathcal{D}$, $x \mapsto \mathbb{E}[\Gamma h(Z) \mid X = x]$ is continuous at c and the path $g_t = g_0 + th$ is admissible for small t . The class $\mathcal{D}$ contains boundary bump directions of the form $h_j(Z) = \varphi(j(X - c))$, where φ is smooth, compactly supported, and $\varphi(0) = 1$.*

Theorem 3 (L^2 non-regularity of threshold welfare). *Under Assumption 3, for every $h \in \mathcal{D}$, the directional derivative of $\Theta_c(g_0 + th)$ at $t = 0$ is*

$$D\Theta_c(g_0)[h] = f_X(c)\mathbb{E}[\Gamma h(Z) \mid X = c].$$

If $f_X(c)\mathbb{E}[\Gamma \mid X = c] \neq 0$, this derivative is not continuous with respect to the $L^2(P_Z)$ norm on $\mathcal{D}$. Hence it has no continuous linear extension to $L^2(P_Z)$ and there is no $\omega \in L^2(P_Z)$ such that

$$D\Theta_c(g_0)[h] = \mathbb{E}[\omega(Z)h(Z)]$$

for all h in the L^2 closure of $\mathcal{D}$.

Proposition 1 (Boundary Hilbert space and Riesz representer). *Let*

$$\mathcal{H}_c = L^2(f_X(c) dP_{Z|X=c}), \quad \langle a, b \rangle_{\mathcal{H}_c} = f_X(c) \mathbb{E}[a(Z)b(Z) \mid X = c].$$

If $\mathbb{E}[\Gamma^2 \mid X = c] < \infty$, then $D\Theta_c(g_0)$ is continuous on $\mathcal{H}_c$ and has Riesz representer $\bar{\Gamma}_c(Z) = \mathbb{E}[\Gamma \mid Z, X = c]$:

$$D\Theta_c(g_0)[h] = \langle \bar{\Gamma}_c, h \rangle_{\mathcal{H}_c}.$$

Equivalently, one may work in the larger space $L^2(f_X(c)dP_{O|X=c})$ and restrict directions to the Z -measurable subspace. Thus the obstruction in Theorem 3 is not the absence of a linear representer in the correct boundary space. It is the fact that ordinary global $L^2(P_Z)$ prediction loss does not control the boundary Hilbert norm.

Proposition 1 gives the positive counterpart of the nonregularity result: boundary-local orthogonalization is meaningful once validation data identify the boundary Hilbert object. Feasible trace estimation therefore uses boundary kernels and incurs the factors $1/h$ and $1/h^2$. Theorem 3 is an L^2 non-regularity result, not an impossibility theorem in every topology. It says that ordinary L^2 first-stage accuracy and ordinary Neyman orthogonalization cannot, by themselves, regularize generated-state threshold welfare; any correction must use boundary-local information, smoothing, or a topology that controls traces. Smooth policies are one way to change the target: for $\pi_{c,b}(x) = \Lambda((x - c)/b)$, the functional is L^2 differentiable for fixed b , but the derivative has norm of order $b^{-1/2}$. The approximation error is of order $b^{1+\alpha}$ under the boundary margin condition because it is bounded by $C\mathbb{E}[|\Gamma|\mathbf{1}\{|X - c| \leq Cb\}]$. Thus smoothing is a regularization device, not a free proof that the sharp-boundary problem is smooth. Boundary information remains necessary: trace, trace-slope, smoothing with an explicit bandwidth, or explicit validation near the cutoff.

At a welfare-maximizing cutoff, the fixed trace can vanish because the first-order condition often implies $E[\Gamma \mid X = c_0] = 0$. The learned-cutoff problem is therefore governed by a trace-slope rather than by the fixed trace alone.

Proposition 2 (Trace-slope non-regularity at an optimal cutoff). *Suppose the conditions of Assumption 3 hold in a neighborhood of c_0 , $m(x) = \mathbb{E}[\Gamma \mid X = x]$ is continuously differentiable, $m(c_0) = 0$, and $\partial_x\{f_X(x)m(x)\}_{x=c_0} \neq 0$. For directions h whose boundary trace is differentiable, define*

$$\mathcal{S}_{c_0}(h) = \partial_c\{f_X(c)\mathbb{E}[\Gamma h(Z) \mid X = c]\}_{c=c_0}.$$

Then $\mathcal{S}_{c_0}$ is not continuous with respect to the global $L^2(P_Z)$ norm on the class of trace directions. Hence, even when the fixed trace is zero at an interior optimum, the first-stage object that enters learned cutoffs is still not controlled by global prediction loss.

3.4 Boundary expansion

Let $h_k(x) = f_X(x)\mathbb{E}[\Gamma\Delta^k \mid X = x]$. The sign convention in this paper is always

$$\Delta = \widehat{X} - X.$$

Assumption 4 (Boundary moment smoothness). *For $k = 1, 2, 3, 4$, the boundary moment functions*

$$h_k(x) = f_X(x)\mathbb{E}[\Gamma\Delta^k \mid X = x]$$

exist in a neighborhood of c and have the derivatives required below, with uniformly bounded fourth-order remainder. The perturbation satisfies $|\Delta| \leq r$ almost surely, or more generally $\mathbb{E}[|\Gamma||\Delta|^4 \mid X = x]f_X(x) = O(r^4)$ uniformly near c , with the corresponding Taylor remainders uniformly integrable. The sign convention is $\Delta = \widehat{X} - X$ throughout.

Theorem 4 (Generated-state boundary expansion). *Under Assumption 4; in particular, when h_1 is continuous at c , h_2 is continuously differentiable at c , h_3 is twice continuously differentiable at c , and the fourth-order boundary remainder is uniformly $O(r^4)$,*

$$\begin{aligned} B_g(c) &= \mathbb{E}[\Gamma\{\mathbf{1}(X + \Delta \geq c) - \mathbf{1}(X \geq c)\}] \\ &= h_1(c) - \frac{1}{2}h'_2(c) + \frac{1}{6}h''_3(c) + O(r^4). \end{aligned}$$

In particular, to second order,

$$B_g(c) = f_X(c)\mathbb{E}[\Gamma\Delta \mid X = c] - \frac{1}{2}\partial_x\{f_X(x)\mathbb{E}[\Gamma\Delta^2 \mid X = x]\}_{x=c} + o(r^2).$$

The first term is a welfare-weighted boundary trace. The second term captures local variance and curvature. The first term can vanish, for example, if $\mathbb{E}[\Gamma\Delta \mid X = c] = 0$, but this is a boundary condition, not a global prediction-loss condition. The sign of the second-order term is economically meaningful: local variance can raise or lower welfare depending on how $f_X(x)\mathbb{E}[\Gamma\Delta^2 \mid X = x]$ changes as one moves through the cutoff.

3.5 Budget-constrained targeting

Many policies treat a fixed share q of the population rather than using an externally fixed cutoff. Let c_q solve $P(X \geq c_q) = q$, and let $c_{g,q}$ solve $P(X + \Delta \geq c_{g,q}) = q$. The following proposition records the boundary implication.

Theorem 5 (Fixed-capacity boundary covariance). *Suppose Assumption 4 holds in a neighborhood of c_q , with $\Delta = O_p(r)$ and the conditional first two moments of Δ smooth at c_q . If $f_X(c_q) > 0$, then*

$$c_{g,q} - c_q = \mathbb{E}[\Delta \mid X = c_q] + O(r^2),$$

and

$$\begin{aligned} & \mathbb{E}[\Gamma\{\mathbf{1}(X + \Delta \geq c_{g,q}) - \mathbf{1}(X \geq c_q)\}] \\ &= f_X(c_q)\text{Cov}(\Gamma, \Delta \mid X = c_q) + O(r^2). \end{aligned}$$

The theorem is economically useful because it separates eligibility cutoffs from fixed-capacity targeting. A common local level shift in the generated state is absorbed by the endogenous cutoff. The first-order welfare loss under a fixed budget is therefore the covariance, at the boundary, between treatment gains and representation error. The statement is local: if the conditional mean of Δ changes with the cutoff, the derivative of that mean enters higher-order terms through the cutoff adjustment.

4 Learning Cutoffs and Calibration

Fixed-policy evaluation depends on a boundary trace. Policy learning depends on a trace-slope. Let $M(c) = \mathbb{E}[\Gamma\mathbf{1}\{X \geq c\}]$. Assume M has a unique maximizer c_0 and

$$M(c_0 + t) - M(c_0) = -\frac{1}{2}\kappa t^2 + o(t^2), \quad \kappa > 0.$$

Let C be a compact cutoff set and let P_n denote the empirical measure over the evaluation sample. The empirical learned cutoff is $\hat{c}_{n,m} \in \arg \max_{c \in C} \hat{M}_{n,m}(c)$, where $\hat{\Gamma}_i$ is the observed or cross-fitted welfare-score estimate and

$$\hat{M}_{n,m}(c) = P_n \hat{\Gamma}_i \mathbf{1}\{\hat{g}_m(Z_i) \geq c\}, \quad M_m(c) = E[\Gamma \mathbf{1}\{\hat{g}_m(Z) \geq c\} \mid \mathcal{V}_m].$$

Here $\hat{g}_m$ is the generated-state map estimated from the validation or calibration sample. The raw numerical cutoff is $\hat{c}_{n,m}$, and the implemented policy is $\hat{\Pi} = \{z : \hat{g}_m(z) \geq \hat{c}_{n,m}\}$. The

latent-equivalent coordinate, $\widetilde{c}_{n,m} = \widehat{c}_{n,m} - a(\widehat{g}_m - g_0)$, is a local coordinate for the induced latent boundary of $\widehat{\Pi}$, not a separately implemented cutoff.

Let the first-stage representation have expansion

$$\widehat{g}_m - g_0 = \eta_m b + \frac{1}{m} \sum_{j=1}^m \ell_g(S_j; \cdot) + r_m,$$

where S_j is the validation sample, b is a deterministic first-stage bias direction, η_m is its scalar local bias sequence, $\ell_g(S_j; \cdot)$ is the first-stage influence function for the generated-state map, and r_m is negligible in the trace-slope norm. Define

$$\mathcal{T}_{c_0}(v) = \left. \frac{\partial}{\partial c} \{f_X(c) \mathbb{E}[\Gamma v(Z) \mid X = c]\} \right|_{c=c_0}.$$

A scalar cutoff can absorb a local common shift in the generated score. Define

$$a(v) = \mathbb{E}[v(Z) \mid X = c_0], \quad \mathcal{T}_{c_0}^\perp(v) = \mathcal{T}_{c_0}(v) - \kappa a(v).$$

The term $\mathcal{T}_{c_0}^\perp$ removes the cutoff-reparameterization direction. If $v(Z) \equiv a$ is a pure location shift, then $\mathcal{T}_{c_0}(v) = \kappa a$ and $\mathcal{T}_{c_0}^\perp(v) = 0$. For the validation influence function below, set

$$\chi^\perp(S_j) = \mathcal{T}_{c_0}^\perp\{\ell_g(S_j; \cdot)\}, \quad \text{Var}(\chi^\perp(S_j)) = \sigma_{T,\perp}^2.$$

Proposition 3 (Cutoff reparameterization invariance). *If $\widehat{X} = X + \theta$ and the cutoff is learned on the same generated-score scale, then the numerical cutoff changes by θ but the induced latent treated set is unchanged:*

$$\mathbf{1}\{X + \theta \geq c_0 + \theta\} = \mathbf{1}\{X \geq c_0\}.$$

Consequently, validation uncertainty in a pure location direction affects the coordinate of the learned cutoff but not policy welfare. Learned-cutoff welfare and regret statements below are therefore stated in the quotient space that removes this common-shift direction, through $\mathcal{T}_{c_0}^\perp$.

For learned-cutoff analysis, the raw fixed-cutoff distortion $B_g(c)$ must be residualized by the local common-shift component. Given a first-stage error δ , write

$$a(\delta) = E[\delta(Z) \mid X = c_0], \quad v_\delta(Z) = \delta(Z) - a(\delta),$$

and define the residualized distortion

$$B_{v_\delta}(c) = E \left[\Gamma \{ \mathbf{1}(X + v_\delta(Z) \geq c) - \mathbf{1}(X \geq c) \} \mid \mathcal{V}_m \right],$$

where $\mathcal{V}_m$ denotes the validation sample. The raw $B_g(c)$ remains the right object for a fixed numerical cutoff. The learned-cutoff drift below uses B_{v_δ} because the raw generated-score cutoff has already shifted by $a(\delta)$.

Assumption 5 (Local cube-root and trace-slope experiment). *For each compact $K \subset \mathbb{R}$, the following hold in $\ell^\infty(K)$ with $c = c_0 + n^{-1/3}u$.*

- (i) *Let $\delta_m(Z) = \widehat{g}_m(Z) - g_0(Z)$, let $a(\delta_m) = E[\delta_m(Z) \mid X = c_0]$ denote the local common-shift component, and set $v_m(Z) = \delta_m(Z) - a(\delta_m)$. The centered local empirical process written in the residualized generated coordinate is*

$$\begin{aligned} \widehat{A}_{n,m}(u) = n^{2/3} (P_n - P) \Gamma \Big[& \mathbf{1}\{X + v_m(Z) \geq c_0 + n^{-1/3}u\} \\ & - \mathbf{1}\{X + v_m(Z) \geq c_0\} \Big]. \end{aligned}$$

This process converges conditionally on the validation sample to the same tight two-sided Gaussian process $\mathbb{G}_\partial(u)$ as the oracle threshold process, and $\mathbb{G}_\partial$ is asymptotically independent of the validation influence function. The covariance kernel is the standard threshold kernel with scale $f_X(c_0)\mathbb{E}[\Gamma^2 \mid X = c_0](|u| + |v| - |u - v|)/2$ on each side of zero.

- (ii) *The population objective satisfies uniformly on K ,*

$$n^{2/3} \{M(c_0 + n^{-1/3}u) - M(c_0)\} \rightarrow -\frac{1}{2}\kappa u^2, \quad \kappa > 0.$$

- (iii) *The residualized generated-state perturbation satisfies the uniform local trace-slope expansion. With $\delta_m = \widehat{g}_m - g_0$, $v_m = v_{\delta_m}$, and B_{v_m} as defined above,*

$$n^{2/3} \left[B_{v_m}(c_0 + n^{-1/3}u) - B_{v_m}(c_0) \right] = n^{1/3} u \mathcal{T}_{c_0}^\perp(\delta_m) + o_p(1).$$

For a pure common shift $\delta_m \equiv a$, the residualized perturbation is zero, so both sides vanish.

- (iv) *Nuisance-score remainders and numerical optimization errors are $o_p(1)$ in the localized*

$n^{2/3}$ objective. Near maximizers are localized at the cube-root rate, as in Lemma 3, and the limiting criterion has an almost surely unique maximizer.

Lemma 2 (Smooth generated-coordinate sufficient conditions). *Let $W_m = X + v_m(Z)$, where $v_m = r_m v_m$ is residualized so that $E[v_m(Z) \mid X = c_0] = 0$, $r_m = O_p(m^{-1/2})$, and $n^{2/3}/m = O(1)$. Work conditionally on the validation sample. Suppose that, in a neighborhood of c_0 , the conditional moments of v_m given $X = x$ are uniformly bounded, and the joint conditional density and the conditional second-moment functions of (X, v_m, Γ) are locally Lipschitz. Suppose also that the class of threshold sets $\{1(W_m \geq c_0 + n^{-1/3}u) : u \in K\}$ is evaluated on a sample split independent of $\mathcal{V}_m$.*

Then, for every fixed $C < \infty$,

$$\sup_{|t| \leq Cn^{-1/3}} |f_{W_m}(c_0 + t) - f_X(c_0)| = o_p(1),$$

and

$$\sup_{|t| \leq Cn^{-1/3}} |E[\Gamma^2 \mid W_m = c_0 + t, \mathcal{V}_m] - E[\Gamma^2 \mid X = c_0]| = o_p(1).$$

Moreover $P(|W_m - c_0| \leq Cn^{-1/3} \mid \mathcal{V}_m) = O_p(n^{-1/3})$. Hence the conditional local empirical process indexed by $\{1(W_m \geq c_0 + n^{-1/3}u) : u \in K\}$ has the same first-order threshold covariance kernel as the oracle process in X . Conditional finite-dimensional convergence and localized VC tightness give stable convergence to $\mathbb{G}_\partial$. Because the validation influence sum is $\mathcal{V}_m$ -measurable and the evaluation fold is split from $\mathcal{V}_m$, this stable limit is independent of the validation Gaussian.

Lemma 3 (Cube-root localization with validation drift). *Suppose the population objective satisfies $M(c_0 + t) - M(c_0) \leq -at^2$ for small t and some $a > 0$. Suppose the threshold empirical process obeys the local maximal inequality*

$$E \left[\sup_{|c - c_0| \leq r} |(P_n - P)\{m_c - m_{c_0}\}| \right] \lesssim n^{-1/2} r^{1/2}$$

for local criterion increments. Suppose the residualized validation drift is $O_p(m^{-1/2})$ with $n^{2/3}/m = O(1)$. Then every near maximizer of the learned-cutoff criterion is $O_p(n^{-1/3})$ after the common-shift coordinate is removed. Validation drift changes the local limit, but not the cube-root localization rate.

Lemma 4 (Cross-fitted nuisance scores in the boundary objective). *Let $\widehat{\Gamma}$ be cross-fitted so*

that, fold by fold, it is conditionally independent of the evaluation empirical process. For u in a compact set, write

$$I_{n,u}(X) = \mathbf{1}\{X \geq c_0 + n^{-1/3}u\} - \mathbf{1}\{X \geq c_0\}.$$

Suppose the boundary tube has probability $O(n^{-1/3})$. Suppose also that, uniformly over fixed compact sets,

$$E[(\widehat{\Gamma} - \Gamma)^2 \mid |X - c_0| \leq Cn^{-1/3}] = o_p(1)$$

and

$$n^{1/3} \sup_{|x-c_0| \leq Cn^{-1/3}} |E[\widehat{\Gamma} - \Gamma \mid X = x]| = o_p(1).$$

For doubly robust scores, the last display follows from the usual orthogonality expansion if the local nuisance product rate in the same boundary tube satisfies

$$n^{1/3} \|\widehat{\mu} - \mu\|_{\text{loc},2} \|\widehat{e} - e\|_{\text{loc},2} = o_p(1),$$

where the local norms are computed on $|X - c_0| \leq Cn^{-1/3}$. A global double-machine-learning product rate alone does not imply this condition; it suffices only when nuisance errors do not concentrate in the $n^{-1/3}$ boundary tube. If the local conditional rates are no worse than global rates, two $n^{-1/4}$ -type nuisance rates satisfy the display. If boundary bias persists, the condition fails and the cube-root criterion is shifted first order. Then

$$\sup_{u \in K} n^{2/3} \left| (P_n - P)\{(\widehat{\Gamma} - \Gamma)I_{n,u}(X)\} \right| = o_p(1),$$

and the population term $\sup_{u \in K} n^{2/3} |P\{(\widehat{\Gamma} - \Gamma)I_{n,u}(X)\}|$ is also $o_p(1)$. Hence replacing Γ by $\widehat{\Gamma}$ does not change the local cube-root experiment.

The smooth generated-coordinate lemma is a sufficient condition for Assumption 5(i), Lemma 3 verifies the localization part of Assumption 5(iv), and Lemma 4 gives a checkable nuisance condition. It can fail if generated-coordinate error does not vanish locally at the boundary, if it creates density spikes near c_0 , or if validation and evaluation samples are not split so that the local empirical process and the validation influence function remain coupled. Together with the curvature, trace-slope, and remainder conditions in Assumption 5(ii)–(iv), it yields the learned-cutoff local experiment. In the calibrated-score framework of Section 4.2, Theorem 9 derives the residualized trace-slope drift rather than assuming it.

Theorem 6 (Learned-cutoff trace-slope limit). *Let $\tilde{c}_{n,m} = \hat{c}_{n,m} - a(\hat{g}_m - g_0)$ denote the latent-equivalent cutoff after removing the local common-shift component. Suppose Assumption 5 holds, $n^{2/3}/m \rightarrow \lambda_g \in [0, \infty)$, and*

$$n^{1/3}\eta_m\mathcal{T}_{c_0}^\perp(b) \rightarrow \beta_b.$$

Then

$$n^{1/3}(\tilde{c}_{n,m} - c_0) \Rightarrow \arg \max_{u \in \mathbb{R}} \left\{ \mathbb{G}_\partial(u) - \frac{1}{2}\kappa u^2 + u(\beta_b + \sqrt{\lambda_g}\sigma_{T,\perp}N_g) \right\},$$

where $N_g \sim N(0, 1)$ is independent of $\mathbb{G}_\partial$.

This high-level theorem isolates the product local experiment. The finite-dimensional calibration-layer theorem below is the primitive implementation used for feasible inference: it verifies the generated-coordinate local process, the residualized trace-slope expansion, and the validation influence function under observable calibration moments. Thus Theorem 6 should be read as the local-experiment representation, while Theorem 9 gives the primitive calibration-layer version.

The theorem is a statement about the policy-relevant latent-equivalent boundary, not about the raw generated-score coordinate. The local empirical process is evaluated in the generated coordinate after subtracting the common shift, while the final normalization is expressed in the latent-equivalent coordinate. The numerical coordinate $\hat{c}_{n,m}$ also contains the common-shift term $a(\hat{g}_m - g_0)$; a pure location shift changes that coordinate but not the induced treated set. The sample-size message concerns the non-reparameterized component. More generally, if $\mathcal{T}_{c_0}^\perp(\hat{g} - g_0) = O_p(r_m)$, then validation uncertainty is first-order negligible only when $n^{1/3}r_m = o(1)$. The displayed $m \asymp n^{2/3}$ frontier is the root- m calibration corollary. If $m \gg n^{2/3}$, the non-reparameterized first-stage trace-slope vanishes from the first-order cutoff distribution. If $m \lesssim n^{2/3}$, ordinary threshold cube-root inference that ignores representation uncertainty is incomplete. A feasible implementation can use a numerical derivative or bootstrap for the evaluation-sample cube-root component together with the validation influence function χ for the trace-slope component; omitting the latter gives the wrong local law at the frontier.

Proposition 4 (Drift-robust and drift-frontier regimes). *In the local experiment of Theorem 6, suppose the calibration layer is regular and root- m . If either $n^{2/3}/m \rightarrow 0$ or $\sigma_{T,\perp} = 0$, validation drift is first-order negligible and the oracle Kim–Pollard law obtains. If $n^{2/3}/m \rightarrow \lambda_g > 0$ and $\sigma_{T,\perp} > 0$, validation drift enters the cutoff law through the*

residualized trace-slope. Pure location shifts that only reparameterize the numerical cutoff satisfy $\mathcal{T}_{c_0}^\perp = 0$. More generally, a first-stage error is drift robust at the frontier only when its residualized trace-slope vanishes; zero fixed-cutoff trace at c_0 alone does not imply this. Sun-type generated-boundary designs impose local common-scale error and payoff/boundary conditional centering. In this notation their drift-robust conclusions correspond to the zero residualized trace-slope face reached by local orthogonality, not by cutoff absorption, or to the faster-first-stage face $n^{2/3}/m \rightarrow 0$. If $n^{2/3}/m \rightarrow \infty$ and $\sigma_{T,\perp} > 0$, the $n^{1/3}$ -normalized cutoff is generally not tight; validation noise dominates, and the appropriate normalization is the $m^{1/2}$ scale for the induced latent-equivalent cutoff shift. The feasible frontier theory focuses on $\lambda_g < \infty$.

Corollary 1 (Same RMSE, different trace-slope). *There exist two generated-state perturbations Δ_1 and Δ_2 with the same global prediction loss, $E[\Delta_1^2] = E[\Delta_2^2]$, but different learned-cutoff first-order laws. Choose Δ_1 supported away from a neighborhood of c_0 , so that its boundary trace and residualized trace-slope vanish. Choose Δ_2 with the same L^2 norm and with*

$$\mathcal{T}_{c_0}^\perp(\Delta_2) \neq 0.$$

A simple sufficient construction is to impose $E[\Delta_2 \mid X = c_0] = 0$ and a nonzero derivative of $f_X(c)E[\Gamma\Delta_2 \mid X = c]$ at c_0 . A global precision criterion treats the two perturbations alike, but a sharp learned cutoff does not. The equality of L^2 losses is global by construction: a boundary-restricted loss would already distinguish the two perturbations, which is precisely why global RMSE is not the welfare-relevant validation target for sharp rules.

Theorem 7 (Rate-sharp validation-frontier lower bound). *Let $\mathcal{P}_{n,m}(C)$ be a product local model class*

$$P_{\alpha,\theta} = P_\alpha^E \otimes P_\theta^V, \quad |\alpha| \leq Cn^{-1/3}, \quad |\theta - \theta_0| \leq Cm^{-1/2},$$

where the evaluation subexperiment has quadratic boundary curvature and the usual cube-root threshold lower bound, the validation subexperiment is LAN with nonsingular information, and the induced latent-equivalent cutoff map satisfies

$$c(\theta_0 + t) = c_0 + \frac{\tau^\perp}{\kappa}t + o(|t|), \quad \tau^\perp = \mathcal{T}_{c_0}^\perp\{\partial_\theta q_\theta(Z)|_{\theta=\theta_0}\} \neq 0.$$

Then every estimator measurable in this product experiment has local minimax regret

bounded below by, for some constant $C' > 0$,

$$\inf_{\bar{c}} \sup_{P \in \mathcal{P}_{n,m}(C)} \mathbb{E}_P\{M_P(c_P) - M_P(\bar{c})\} \geq C' \max\{n^{-2/3}, m^{-1}\}.$$

Consequently the rate frontier $m \asymp n^{2/3}$ is not a proof artifact: $m \gg n^{2/3}$ is necessary for validation error to be asymptotically negligible relative to ordinary threshold-learning regret. The theorem is a rate lower bound, not a constant-sharp minimax equality. It gives necessity of the two rates; attainability at the frontier is supplied by the local limit experiment in Theorem 6, together with Corollary 3.

Remark 2 (Total budget and non-split samples). *The max lower bound has a budget interpretation. Evaluation noise contributes the ordinary threshold regret rate $n^{-2/3}$. Root- m validation contributes m^{-1} through the squared trace-slope drift. Equalizing these rates gives $m \asymp n^{2/3}$. Without sample splitting, the rate frontier is unchanged, but the local product experiment need not have independent components. For example, in the oracle coordinate let*

$$\ell_{n,u}(O) = \Gamma\{\mathbf{1}(X \geq c_0 + n^{-1/3}u) - \mathbf{1}(X \geq c_0)\}.$$

If

$$G_{n,\partial}(u) = n^{2/3}(P_n - P)\ell_{n,u}(O), \quad N_{g,n} = m^{-1/2} \sum_{j=1}^m \chi_j^\perp / \sigma_{T,\perp},$$

then a non-split design can have

$$\text{Cov}\{\mathbb{G}_\partial(u), N_g\} = \lim \text{Cov} \left[n^{2/3}(P_n - P)\ell_{n,u}(O), m^{-1/2} \sum_{j=1}^m \chi_j^\perp / \sigma_{T,\perp} \right].$$

The additive variance formula and the feasible simulations in the supplement use split-sample independence; without it, this covariance must be estimated or removed by cross-fitting.

Corollary 2 (Additive variance under the canonical threshold process). *Let $\mathbb{G}_\partial$ be the canonical two-sided threshold Gaussian process with stationary increments and covariance kernel proportional to $(|u| + |v| - |u - v|)/2$. Let $\xi = \beta_b + \sqrt{\lambda_g} \sigma_{T,\perp} N_g$ be independent of $\mathbb{G}_\partial$. Then*

$$\arg \max_u \{\mathbb{G}_\partial(u) - \kappa u^2/2 + u\xi\} \stackrel{d}{=} \zeta_0 + \xi/\kappa, \quad \zeta_0 = \arg \max_u \{\mathbb{G}_\partial(u) - \kappa u^2/2\}.$$

Thus the normalized cutoff variance is

$$\text{Var}(\zeta_0) + \lambda_g \sigma_{T,\perp}^2 / \kappa^2.$$

This is a distributional identity, not a pathwise completion-of-squares argument. It uses stationary increments of the two-sided threshold process and split-sample independence. Without those conditions, the argmax representation remains the correct statement, but the additive variance formula need not hold. Recent efficiency work studies regret distributions for learned policies more generally (Fang et al., 2025); the extra component here comes from validation of a generated boundary state.

4.1 A primitive heterogeneous trace example

This section gives a worked example in which the trace-slope is visible in closed form. The example verifies the high-level objects in Theorem 6 in a primitive data-generating process and motivates the heterogeneous generated-state perturbation used in the simulation.

Let

$$X \sim U[-1, 1], \quad G \in \{-1, +1\}, \quad \mathbb{P}(G = 1) = \mathbb{P}(G = -1) = 1/2,$$

with X , G , and ε mutually independent, $\mathbb{E}[\varepsilon] = 0$, and $\text{Var}(\varepsilon) = \sigma_\varepsilon^2$. The welfare score is

$$\Gamma = X(1 + \lambda G) + \varepsilon, \quad \lambda \neq 0.$$

The latent policy state is X . The generated state is $\widehat{X} = X + \widehat{\theta}G$, where $\widehat{\theta}$ is estimated from an independent validation sample and satisfies

$$\sqrt{m} \widehat{\theta} \Rightarrow N(0, \sigma_\theta^2).$$

For a fixed scalar cutoff, the policy class is $\pi_c(x) = \mathbf{1}\{x \geq c\}$.

This perturbation is not a scalar reparameterization of the cutoff. A common shift $X + \widehat{\theta}$ can be absorbed by replacing c with $c - \widehat{\theta}$. By contrast, $X + \widehat{\theta}G$ shifts the two groups in opposite directions. A single scalar cutoff cannot remove the representation error for both groups.

Proposition 5 (Primitive heterogeneous trace and trace-slope). *In the heterogeneous-trace*

model above, the latent-state objective

$$M_0(c) = \mathbb{E}[\Gamma \mathbf{1}\{X \geq c\}]$$

has unique maximizer $c_0 = 0$ and local curvature $\kappa = 1/2$, so that

$$M_0(c) - M_0(0) = -\frac{1}{4}c^2 + o(c^2) = -\frac{1}{2}\kappa c^2 + o(c^2).$$

Let the perturbation direction be $v(Z) = G$. Then the fixed-policy trace at the optimal cutoff is zero:

$$f_X(0)\mathbb{E}[\Gamma G \mid X = 0] = 0.$$

However, the trace-slope is nonzero:

$$\mathcal{T}_0(G) = \left. \frac{\partial}{\partial c} \{f_X(c)\mathbb{E}[\Gamma G \mid X = c]\} \right|_{c=0} = \frac{\lambda}{2}.$$

Consequently, if $n^{2/3}/m \rightarrow \lambda_g$, then the generated-state term in the localized learned-cutoff criterion is

$$n^{1/3}u \mathcal{T}_0(\widehat{\theta}G) \Rightarrow u \frac{\lambda}{2} \sqrt{\lambda_g} \sigma_\theta N_g.$$

Equivalently, because $\kappa = 1/2$, the first-stage shift in the local cutoff distribution has coefficient

$$\frac{\mathcal{T}_0(G)}{\kappa} = \lambda.$$

Because $E[G \mid X = 0] = 0$, the local common-shift component is $a(\widehat{\theta}G) = 0$, so the raw and residualized perturbations coincide in this example. Conditional on the validation draw $\widehat{\theta}$, the generated running variable is $W_m = X + \widehat{\theta}G$. For $|\widehat{\theta}| = o_p(1)$ and local $|t| \leq Cn^{-1/3}$, the density of W_m at t is $1/2 + O_p(|\widehat{\theta}| + n^{-1/3})$ away from the endpoints of $[-1, 1]$. The induced selection on G changes $E[\Gamma^2 \mid W_m = t, \widehat{\theta}]$ by $O_p(|\widehat{\theta}| + n^{-1/3})$. At the frontier $|\widehat{\theta}| = O_p(m^{-1/2}) = O_p(n^{-1/3})$, so this perturbation is $o_p(1)$ for the first-order threshold covariance kernel:

$$E[\Gamma^2 \mid W_m = t, \widehat{\theta}] \rightarrow E[\Gamma^2 \mid X = 0].$$

The threshold class in W_m is one-dimensional VC conditional on the validation sample. Hence the conditional local empirical process satisfies Lemma 2; by sample splitting its two-sided Gaussian limit is independent of the validation Gaussian limit N_g . A nonuniform-density

variant in the supplement verifies the same conditions when the local density has curvature, so the argument is not an artifact of a flat running-variable density. If the evaluation-sample local argmax is denoted by ζ_0 , independent of the validation limit, then

$$n^{1/3}(\widehat{c} - c_0) \Rightarrow \zeta_0 + \lambda\sqrt{\lambda_g}\sigma_\theta N_g.$$

Thus

$$\text{Var}\{n^{1/3}(\widehat{c} - c_0)\} \rightarrow \text{Var}(\zeta_0) + \lambda^2\lambda_g\sigma_\theta^2.$$

At $c_0 = 0$, the generated-state perturbation has no first-order effect on fixed-policy welfare because the welfare-weighted group imbalance vanishes exactly at the boundary. Policy learning, however, moves the cutoff locally. The relevant object is therefore the derivative of the trace as the boundary moves. That derivative is $\lambda/2$, so the learned cutoff remains sensitive to validation error. The learned-cutoff theorem is thus not a fixed-policy boundary expansion with an argmax applied afterward.

The example also explains the variance decomposition reported in the simulations. With $\lambda = 1$ and $\widehat{\theta} \sim N(0, 1/m)$,

$$\text{Var}\{n^{1/3}(\widehat{c} - c_0)\} \approx \text{Var}\{n^{1/3}(\widehat{c}_{\text{oracle}} - c_0)\} + \text{Var}(n^{1/3}\widehat{\theta}).$$

The simulation implements this design directly and verifies the phase transition across $m = n^a$.

4.2 A primitive calibrated generated-state model

The trace and trace-slope are not merely high-level objects. They are estimable in a common generated-state pipeline. This section gives a primitive semiparametric model in which the boundary trace has an ordinary validation-sample influence function even though the welfare functional itself is nonregular in the representation map.

Let $S_1, \dots, S_m$ be an independent validation or calibration sample. The generated state is

$$\widehat{X} = q_{\widehat{\theta}}(Z), \quad X = q_{\theta_0}(Z), \quad \theta_0 \in \Theta \subset \mathbb{R}^d,$$

where q_θ is a finite-dimensional calibration layer on top of a frozen upstream representation, such as a fixed text embedding, factor score, or risk score. The theorem below is not an end-to-end representation-learning theorem; upstream uncertainty must be included in the calibration influence function or diagnosed near the boundary. The calibration parameter

solves

$$\mathbb{E}[\psi(S; \theta_0)] = 0, \quad \frac{1}{m} \sum_{j=1}^m \psi(S_j; \hat{\theta}) = 0.$$

Let

$$H = \mathbb{E}[\partial_\theta \psi(S; \theta_0)], \quad \Omega = \mathbb{E}[\psi(S; \theta_0) \psi(S; \theta_0)^\top],$$

and let $\dot{q}_{\theta_0}(Z) = \partial_\theta q_\theta(Z)|_{\theta=\theta_0}$, a d -vector. Define the calibrated boundary trace vector, its derivative, and the common-shift loading vector

$$A(c) = f_X(c) \mathbb{E}[\Gamma \dot{q}_{\theta_0}(Z) \mid X = c],$$

$$\dot{A}(c) = \partial_c \{f_X(c) \mathbb{E}[\Gamma \dot{q}_{\theta_0}(Z) \mid X = c]\}, \quad D(c) = \mathbb{E}[\dot{q}_{\theta_0}(Z) \mid X = c].$$

The non-reparameterized trace-slope loading at the optimum is

$$\dot{A}_\perp(c_0) = \dot{A}(c_0) - \kappa D(c_0).$$

The high-level fixed-policy trace is $A(c)^\top (\hat{\theta} - \theta_0)$, while the high-level learned-cutoff trace-slope that affects the policy-equivalent cutoff is $\dot{A}_\perp(c_0)^\top (\hat{\theta} - \theta_0)$.

Assumption 6 (Primitive calibrated generated score). *(i) θ_0 is an interior point of Θ , H is nonsingular, $\mathbb{E} \|\psi(S; \theta_0)\|^{2+\delta} < \infty$ for some $\delta > 0$, and the estimating equation is locally differentiable with a stochastic equicontinuity remainder small enough for the usual Z -estimator expansion.*

- (ii) $q_\theta(z)$ is twice continuously differentiable in a neighborhood of θ_0 , with derivatives dominated by an integrable envelope. The map $x \mapsto f_X(x) \mathbb{E}[\Gamma \dot{q}_{\theta_0}(Z) \mid X = x]$ is continuously differentiable in a neighborhood of the cutoff c_0 .*
- (iii) Evaluation and validation samples are independent in the main product experiment. K -fold cross-fitting is admissible only when each fold's local evaluation criterion uses first-stage and nuisance objects estimated on observations excluded from that fold; otherwise the limit experiment may contain an additional covariance term. The nuisance score satisfies the fixed-policy trace remainder condition for trace inference and the localized boundary-tube condition in Lemma 4 for learned-cutoff inference.*
- (iv) For feasible kernel estimation, K is compactly supported, twice continuously differentiable, integrates to one, has bounded derivatives through order two, and has enough*

moment cancellations for the stated bias order. The bandwidth $h \downarrow 0$ satisfies the restrictions stated in Theorem 10.

Theorem 8 (Primitive influence functions for trace and trace-slope). *Under Assumption 6,*

$$\widehat{\theta} - \theta_0 = -H^{-1} \frac{1}{m} \sum_{j=1}^m \psi(S_j; \theta_0) + o_p(m^{-1/2}).$$

For a fixed threshold c ,

$$B_m(c) := \mathbb{E}[\Gamma\{\mathbf{1}(q_{\widehat{\theta}}(Z) \geq c) - \mathbf{1}(q_{\theta_0}(Z) \geq c)\} \mid \widehat{\theta}]$$

satisfies

$$B_m(c) = A(c)^\top (\widehat{\theta} - \theta_0) + o_p(m^{-1/2}) = \frac{1}{m} \sum_{j=1}^m b_c(S_j) + o_p(m^{-1/2}),$$

where

$$b_c(S) = -A(c)^\top H^{-1} \psi(S; \theta_0).$$

Consequently,

$$\sqrt{m} B_m(c) \Rightarrow N(0, \sigma_c^2), \quad \sigma_c^2 = A(c)^\top H^{-1} \Omega H^{-\top} A(c).$$

For learned cutoff problems, the non-reparameterized trace-slope satisfies

$$\mathcal{T}_{c_0}^\perp(\widehat{g} - g_0) = \dot{A}_\perp(c_0)^\top (\widehat{\theta} - \theta_0) + o_p(m^{-1/2}) = \frac{1}{m} \sum_{j=1}^m \chi_{c_0}^\perp(S_j) + o_p(m^{-1/2}),$$

where

$$\chi_{c_0}^\perp(S) = -\dot{A}_\perp(c_0)^\top H^{-1} \psi(S; \theta_0),$$

and

$$\sqrt{m} \mathcal{T}_{c_0}^\perp(\widehat{g} - g_0) \Rightarrow N(0, \sigma_{T,\perp}^2), \quad \sigma_{T,\perp}^2 = \dot{A}_\perp(c_0)^\top H^{-1} \Omega H^{-\top} \dot{A}_\perp(c_0).$$

The variance formulas condition on the cross-fitted evaluation sample used to estimate the welfare score. Under sample splitting, nuisance estimation in $\widehat{\Gamma}$ contributes through the empirical-process remainder and not through the validation variance σ_c^2 or $\sigma_{T,\perp}^2$.

Thus, if $n^{2/3}/m \rightarrow \lambda_g$, the first-stage component in Theorem 6 contributes the Gaussian

drift

$$\sqrt{\lambda_g} \sigma_{T,\perp} N_g.$$

Theorem 9 (Calibrated primitive drift for the learned-cutoff experiment). *Suppose the generated-coordinate empirical process, curvature, and nuisance conditions in Assumption 5(i),(ii),(iv) hold. Suppose also that Assumption 6 holds. Let*

$$\delta_{\hat{\theta}}(Z) = q_{\hat{\theta}}(Z) - q_{\theta_0}(Z), \quad v_{\hat{\theta}}(Z) = \delta_{\hat{\theta}}(Z) - a(\delta_{\hat{\theta}}).$$

If q_{θ} and the conditional law of $(X, \dot{q}_{\theta_0}(Z), \Gamma)$ are locally smooth enough that the density and second-moment conditions in Lemma 2 hold for $W_m = X + v_{\hat{\theta}}(Z)$, then, uniformly for u in compact sets,

$$n^{2/3} \{B_{v_{\hat{\theta}}}(c_0 + n^{-1/3}u) - B_{v_{\hat{\theta}}}(c_0)\} = n^{1/3}u \mathcal{T}_{c_0}^{\perp}(q_{\hat{\theta}} - q_{\theta_0}) + o_p(1).$$

Moreover,

$$\mathcal{T}_{c_0}^{\perp}(q_{\hat{\theta}} - q_{\theta_0}) = \dot{A}_{\perp}(c_0)^{\top}(\hat{\theta} - \theta_0) + o_p(m^{-1/2}),$$

where $\dot{A}_{\perp}(c_0)$ subtracts the cutoff-reparameterization loading. Hence Theorem 6 applies with $\sigma_{T,\perp}^2 = \dot{A}_{\perp}(c_0)^{\top} H^{-1} \Omega H^{-\top} \dot{A}_{\perp}(c_0)$.

Corollary 3 (Primitive validation frontier under calibrated generated states). *Suppose $\hat{X} = q_{\hat{\theta}}(Z)$ and $X = q_{\theta_0}(Z)$ are generated by the calibrated finite-dimensional model in Assumption 6. Suppose the honest-split design, curvature, smooth generated-coordinate, and localized nuisance conditions above hold, and suppose $n^{2/3}/m \rightarrow \lambda_g < \infty$. Then the latent-equivalent cutoff obeys*

$$n^{1/3}(\tilde{c}_{n,m} - c_0) \Rightarrow \arg \max_{u \in \mathbb{R}} \left\{ \mathbb{G}_{\partial}(u) - \frac{1}{2} \kappa u^2 + u(\beta_b + \sqrt{\lambda_g} \sigma_{T,\perp} N_g) \right\},$$

with $N_g \sim N(0, 1)$ independent of $\mathbb{G}_{\partial}$. This statement uses the honest-split product experiment. Non-split versions require estimating or removing the evaluation-validation covariance described in Remark 2. In the correctly specified centered calibration model, $\beta_b = 0$. This corollary is the primitive version of the high-level local experiment.

The theorem gives an estimable interpretation to the abstract trace-slope. The object that enters the cube-root local experiment is not the validation-sample RMSE of the generated state. It is the validation influence function projected onto the derivative of the welfare-weighted

boundary trace. The variance formulas condition on the cross-fitted welfare score; estimation error in the nuisance components of $\widehat{\Gamma}$ is controlled by the localized boundary-tube condition in Assumption 6(iii) and Lemma 4, and is not part of the validation-sample sandwich variance. This is a θ -recovery result for the calibration layer. If the upstream representation is retrained, its uncertainty must either be included in ψ or diagnosed separately; a practical validation check is whether residual calibration error, after projecting on $\dot{q}_{\widehat{\theta}}$, has nonzero conditional mean near the cutoff.

Proposition 6 (Modular upstream influence functions). *A sufficient boundary-relevant seminorm for an upstream remainder r is*

$$|r|_{\partial} = \sup_{|t| \leq Cn^{-1/3}} \mathbb{E}[|r(Z)| \mid X = c_0 + t] + |D_r(c_0)| + |\dot{A}_{\perp,r}(c_0)|.$$

Here $D_r(c_0) = E[r(Z) \mid X = c_0]$ is the common-shift loading of the remainder, and $\dot{A}_{\perp,r}(c_0)$ is its residualized trace-slope loading, obtained by replacing $\dot{q}_{\theta_0}$ with r in the definition of $\dot{A}_{\perp}$. The seminorm controls the boundary-tube remainder, the common-shift loading, and the residualized trace-slope loading used by the local experiment. Suppose the generated state can be written as $q_{\theta,\rho}(Z)$, where ρ indexes an upstream representation. If the joint first-stage estimator satisfies

$$\begin{pmatrix} \widehat{\theta} - \theta_0 \\ \widehat{\rho} - \rho_0 \end{pmatrix} = \frac{1}{m} \sum_{j=1}^m \Phi(S_j) + o_p(m^{-1/2})$$

in the boundary-relevant norm, then the calibration-layer formulas remain valid after replacing $\dot{q}_{\theta_0}$ by the full derivative $(\partial_{\theta} q_{\theta_0,\rho_0}, \partial_{\rho} q_{\theta_0,\rho_0})$ and replacing ψ by the augmented influence function Φ . If a retrained upstream representation has a non-negligible nonlinear remainder inside the boundary tube, this modular result does not apply.

Theorem 10 (Feasible trace and trace-slope variance estimation). *Suppose Assumption 6 holds and let $s \geq 2$ denote the local smoothness order of $x \mapsto f_X(x) \mathbb{E}[\Gamma \dot{q}_{\theta_0}(Z) \mid X = x]$. Define*

$$\begin{aligned} \widehat{A}(c) &= \frac{1}{nh} \sum_{i=1}^n \widehat{\Gamma}_i \dot{q}_{\widehat{\theta}}(Z_i) K\left(\frac{\widehat{X}_i - c}{h}\right), \\ \widehat{\hat{A}}(c) &= -\frac{1}{nh^2} \sum_{i=1}^n \widehat{\Gamma}_i \dot{q}_{\widehat{\theta}}(Z_i) K'\left(\frac{\widehat{X}_i - c}{h}\right). \end{aligned}$$

Let

$$\widehat{H} = \frac{1}{m} \sum_{j=1}^m \partial_{\theta} \psi(S_j; \widehat{\theta}), \quad \widehat{\Omega} = \frac{1}{m} \sum_{j=1}^m \psi(S_j; \widehat{\theta}) \psi(S_j; \widehat{\theta})^{\top}.$$

If

$$h^s + (nh)^{-1/2} + m^{-1/2}/h \rightarrow 0,$$

then $\widehat{A}(c) \rightarrow_p A(c)$ and

$$\widehat{\sigma}_c^2 := \widehat{A}(c)^{\top} \widehat{H}^{-1} \widehat{\Omega} \widehat{H}^{-\top} \widehat{A}(c) \rightarrow_p \sigma_c^2.$$

If, in addition,

$$h^{s-1} + (nh^3)^{-1/2} + m^{-1/2}/h^2 \rightarrow 0,$$

then $\widehat{\dot{A}}(c_0) \rightarrow_p \dot{A}(c_0)$. If $\widehat{\kappa} \rightarrow_p \kappa$ and

$$\widehat{D}(c_0) = \frac{\sum_{i=1}^n \dot{q}_{\widehat{\theta}}(Z_i) K((\widehat{X}_i - c_0)/h)}{\sum_{i=1}^n K((\widehat{X}_i - c_0)/h)} \rightarrow_p D(c_0),$$

then $\widehat{\dot{A}}_{\perp}(c_0) = \widehat{\dot{A}}(c_0) - \widehat{\kappa} \widehat{D}(c_0) \rightarrow_p \dot{A}_{\perp}(c_0)$ and

$$\widehat{\sigma}_{T,\perp}^2 := \widehat{\dot{A}}_{\perp}(c_0)^{\top} \widehat{H}^{-1} \widehat{\Omega} \widehat{H}^{-\top} \widehat{\dot{A}}_{\perp}(c_0) \rightarrow_p \sigma_{T,\perp}^2.$$

Corollary 4 (Plug-in cutoff for trace-slope variance). *In Theorem 10, let $\widehat{c}$ be a preliminary cutoff estimator. If $|\widehat{c} - c_0|/h = o_p(1)$ and the local derivative estimator is stochastically equicontinuous over $|c - c_0| \leq |\widehat{c} - c_0|$, then $\widehat{\dot{A}}(\widehat{c}) - \widehat{\dot{A}}(c_0) = o_p(1)$ and the same plug-in variance estimator is consistent with $\widehat{c}$ in place of c_0 . At the frontier, $|\widehat{c} - c_0| = O_p(n^{-1/3})$ and the feasible bandwidths in Remark 4 satisfy $n^{-1/3}/h = o(1)$.*

Reparameterization invariance and feasible targets. The local coordinate $\widetilde{c}_{n,m}$ is an analytical coordinate for the latent boundary induced by the implemented set $\widehat{\Pi} = \{z : \widehat{g}_m(z) \geq \widehat{c}_{n,m}\}$. It removes the local common shift of the generated score. That common shift changes the raw numerical cutoff but not the induced latent selection set, and it is not a separately observable target from a single sample. The invariant policy object is the welfare loss of the implemented boundary. If

$$\zeta_{n,m} = n^{1/3}(\widetilde{c}_{n,m} - c_0), \quad R_{n,m} = M(c_0) - M(\widetilde{c}_{n,m}),$$

then local quadratic curvature gives

$$n^{2/3}R_{n,m} = \frac{\kappa}{2}\zeta_{n,m}^2 + o_p(1).$$

Thus feasible inference is reported for the policy-relevant local regret or welfare loss. A raw generated-score cutoff interval is possible, but it must also include the common-shift channel in Remark 3.

Algorithm 1: validation-adjusted local regret inference. The implementation has four steps.

1. Estimate the learned cutoff and $\widehat{\kappa}$, $\widehat{A}_\perp$, and $\widehat{\sigma}_{T,\perp}$.
2. Build $\widehat{\mathbb{G}}_\partial^*(u)$ by reshaped cube-root resampling, or by an equivalent local approximation.
3. Draw $N^* \sim N(0, 1)$ independently and form the simulated criterion in Proposition 7.
4. Let $\widehat{\zeta}^*$ be the simulated argmax and report

$$\widehat{R}_{\text{loc}}^* = \frac{1}{2}\widehat{\kappa}(\widehat{\zeta}^*)^2,$$

with scale $n^{-2/3}\widehat{R}_{\text{loc}}^*$ in welfare units.

The raw generated-score cutoff also contains the common shift and is not the invariant target of the validation-frontier law.

Proposition 7 (Feasible local regret inference for learned cutoffs). *Suppose Theorem 6 holds and the estimators in Theorem 10 are consistent. Let $\widehat{\kappa}$, $\widehat{\lambda}_g$, and $\widehat{\sigma}_{T,\perp}$ be consistent for κ , $n^{2/3}/m$, and the non-reparameterized trace-slope standard deviation. Suppose either the deterministic first-stage trace-slope bias is negligible, $\beta_b = 0$, or a consistent centering estimate $\widehat{\beta}_b$ is available; in the negligible-bias case set $\widehat{\beta}_b = 0$. If the reshaped evaluation process in Algorithm 1 consistently approximates the law of $\mathbb{G}_\partial(u) - \kappa u^2/2$, then the simulated local criterion*

$$\widehat{Q}^*(u) = \widehat{\mathbb{G}}_\partial^*(u) - \frac{1}{2}\widehat{\kappa}u^2 + u\{\widehat{\beta}_b + \sqrt{\widehat{\lambda}_g}\widehat{\sigma}_{T,\perp}N^*\}$$

with $N^* \sim N(0, 1)$ independent of the evaluation-sample resampling consistently approximates the analytical local coordinate $\zeta_{n,m} = n^{1/3}(\bar{c}_{n,m} - c_0)$ in the product experiment. Consequently,

$$\widehat{R}_{\text{loc}}^* = \frac{1}{2}\widehat{\kappa}(\widehat{\zeta}^*)^2, \quad \widehat{\zeta}^* \in \arg \max_u \widehat{Q}^*(u),$$

consistently approximates the law of the invariant local regret $n^{2/3}R_{n,m}$. If $m/n^{2/3} \rightarrow \infty$,

the validation term drops out; at the frontier it must be retained.

Remark 3 (Raw generated-score cutoff law). *A raw numerical cutoff interval must add back the common-shift channel. Let Z_θ denote the Gaussian limit of $\sqrt{m}(\hat{\theta} - \theta_0)$, with covariance $H^{-1}\Omega H^{-\top}$. If*

$$a(\hat{g}_m - g_0) = D(c_0)^\top (\hat{\theta} - \theta_0) + o_p(m^{-1/2}),$$

then the raw generated-score coordinate satisfies

$$n^{1/3}(\hat{c}_{n,m} - c_0) \Rightarrow \arg \max_u \left\{ \mathbb{G}_\partial(u) - \frac{1}{2}\kappa u^2 + u(\beta_b + \sqrt{\lambda_g} \dot{A}_\perp(c_0)^\top Z_\theta) \right\} + \sqrt{\lambda_g} D(c_0)^\top Z_\theta.$$

The two validation channels are generally correlated because they are functions of the same calibration influence vector. This raw-coordinate law is coordinate dependent. The main feasible output above is the policy-invariant local regret.

Corollary 5 (Bias-aware local regret inference). *Under Proposition 7, suppose the deterministic drift belongs to a known interval $\beta_b \in \mathcal{B} = [-B, B]$. Let $C_\alpha^R(\beta)$ be the simulated local-regret set obtained from the criterion with linear drift $u\beta$. Then*

$$C_\alpha^{R,BA} = \bigcup_{\beta \in \mathcal{B}} C_\alpha^R(\beta)$$

has asymptotic coverage at least $1 - \alpha$ uniformly over $\beta_b \in \mathcal{B}$. The reason is direct: for the true β_b , the nominal regret set $C_\alpha^R(\beta_b)$ is contained in the displayed union.

The derivative estimator is more demanding than the trace estimator. Fixed-policy inference only requires estimating a boundary density-weighted conditional moment. Learned-cutoff inference requires estimating its derivative with respect to the cutoff. This distinction is the feasible counterpart of the theoretical distinction between trace and trace-slope. The proposition separates three bias regimes. In the correctly specified primitive calibration model, $\beta_b = 0$. Nonzero β_b is retained for regularized or biased first-stage procedures and is handled as a sensitivity parameter through the bias-aware local-regret interval unless a structural bias model is imposed. In an estimable-bias regime, a consistent $\hat{\beta}_b$ can be used. If only a bound $\beta_b \in [-B, B]$ is credible, a bias-aware local-regret interval is obtained by simulating the displayed criterion over $\beta \in [-B, B]$ and taking the union of the resulting local-regret quantiles. The paper's feasible inference statements use the centered or estimable-bias regime; the bias-aware local-regret version is a conservative sensitivity check.

Remark 4 (Feasible bandwidths at the learned-cutoff frontier). *The derivative condition can*

be read as

$$h \rightarrow 0, \quad nh^3 \rightarrow \infty, \quad mh^4 \rightarrow \infty, \quad h^{s-1} \rightarrow 0.$$

At the critical frontier $m \asymp n^{2/3}$, a bandwidth $h = n^{-\xi}$ is feasible whenever $0 < \xi < 1/6$: the derivative term $(nh^3)^{-1/2}$ requires $\xi < 1/3$, while the generated-state replacement term $m^{-1/2}/h^2 = n^{-1/3+2\xi}$ requires $\xi < 1/6$, which is binding. Thus feasible trace-slope variance estimation is possible at the frontier, but it requires more smoothing than fixed-policy trace estimation.

The supplemental appendix examines the plug-in local simulation behind Proposition 7. The implementation is conservative in the reported design. At $m = n^{2/3}$, the validation-omitted local-coordinate diagnostic covers about 0.62 of simulated draws, while the validation-adjusted diagnostic covers about 0.99. These calculations diagnose omitted validation drift and plug-in conservatism; Algorithm 1 reports the policy-invariant induced local-regret law. The supplement separates the oracle-scale component from plug-in curvature and bandwidth conservatism. A cross-fitted score simulation also shows that well-specified nuisance estimates are asymptotically negligible, whereas boundary bias is first order.

Semi-synthetic boundary exercise. The supplemental appendix gives an empirically calibrated semi-synthetic illustration of the validation-frontier calculation. It uses the Alatas panel only to estimate the empirical boundary shape at the baseline capacity, then constructs two artificial generated-state errors with the same global RMSE. The near-orthogonal perturbation has a small residualized trace-slope; the boundary-local perturbation has a much larger trace-slope. At $n = 10,000$ and $m = n^{2/3}$, the boundary-local perturbation also has a larger cutoff variance. The exercise is semi-synthetic and is not presented as a new empirical application.

Perturbation	Global RMSE	$ \widehat{\mathcal{T}}_{\perp} $	Frontier cutoff var.
Δ_A (near-orthogonal)	1.000	0.036	0.94
Δ_B (boundary-local)	1.000	0.232	2.66

Equal global prediction loss does not imply equal boundary trace-slope, or equal learned-cutoff law.

Alatas diagnostic. The supplement also reports a boundary-covariance computation in the Indonesia poverty-targeting data of Alatas et al. (2012). The calculation is descriptive. It shows the fixed-capacity calculation and reports scope limitations of the approximation.

5 Conclusion

Sharp targeting rules built on calibrated generated states raise a boundary problem that ordinary prediction loss, smooth generated-regressor theory, and standard orthogonal-score arguments do not capture. Fixed-policy welfare depends on a boundary trace. Fixed-capacity welfare depends on a boundary covariance. Learned-policy selection depends on a non-reparameterized trace-slope. Root- m validation data face the frontier $m \asymp n^{2/3}$. The relevant object is not global prediction loss, but the boundary-local trace, covariance, or trace-slope that matches the policy rule.

The primitive theory gives a complete first-order account for fixed representations with a finite-dimensional calibration layer. Retrained representations enter when the upstream step has a boundary-local influence expansion. Sharper finite-sample inference for derivative trace-slope variance remains an implementation problem. In applications, generated scores should be compared by their boundary trace-slope, not only by global RMSE.

A Auxiliary Lemmas

The proofs use four recurring devices. Boundary-tube rearrangement is stated earlier in Lemma 1; the remaining three auxiliary lemmas below give the signed boundary-crossing expansion, localized argmax transfer, and kernel replacement for generated states inside boundary kernels. Together they keep the trace, learned-cutoff, calibrated-score, and optional-extension proofs aligned.

Lemma 5 (Signed boundary-crossing expansion). *Let d denote a realization of Δ . Suppose that the signed conditional measure*

$$\mu_x(\mathrm{d}d) = f_X(x)E[\Gamma \mathbf{1}\{\Delta \in \mathrm{d}d\} \mid X = x]$$

is twice differentiable in x in a neighborhood of c , with derivatives that are dominated by integrable functions of d , and that its third-order Taylor remainder satisfies

$$\int |d| \sup_{0 \leq t \leq 1} \left| \mu_{c-td}(\mathrm{d}d) - \mu_c(\mathrm{d}d) + td \partial_x \mu_c(\mathrm{d}d) - \frac{t^2 d^2}{2} \partial_x^2 \mu_c(\mathrm{d}d) \right| = O(r^4).$$

Equivalently, in terms of the boundary moments $h_k(x) = f_X(x)E[\Gamma \Delta^k \mid X = x]$, assume h_1 is continuous, h_2 is differentiable, h_3 is twice differentiable at c , and the displayed

signed-interval remainder is $O(r^4)$. Then

$$E[\Gamma\{1(X + \Delta \geq c) - 1(X \geq c)\}] = h_1(c) - \frac{1}{2}h'_2(c) + \frac{1}{6}h''_3(c) + O(r^4).$$

Proof. For a fixed value d , the discontinuous difference is exactly the signed interval that crosses the cutoff:

$$1\{x + d \geq c\} - 1\{x \geq c\} = 1\{d > 0\}1\{x \in [c - d, c)\} - 1\{d < 0\}1\{x \in [c, c - d)\}.$$

Using oriented integrals, this gives the identity

$$E[\Gamma\{1(X + \Delta \geq c) - 1(X \geq c)\}] = \int \int_{c-d}^c \mu_x(\mathrm{d}d) \mathrm{d}x = \int d \int_0^1 \mu_{c-t d}(\mathrm{d}d) \mathrm{d}t.$$

Taylor expand the signed measure $x \mapsto \mu_x(\mathrm{d}d)$ at c , integrate over t , and then over d :

$$\begin{aligned} \int d \int_0^1 \mu_{c-t d}(\mathrm{d}d) \mathrm{d}t &= \int d \mu_c(\mathrm{d}d) - \frac{1}{2} \int d^2 \partial_x \mu_c(\mathrm{d}d) + \frac{1}{6} \int d^3 \partial_x^2 \mu_c(\mathrm{d}d) + O(r^4) \\ &= h_1(c) - \frac{1}{2}h'_2(c) + \frac{1}{6}h''_3(c) + O(r^4). \end{aligned}$$

This proof never differentiates a mollified indicator. The remainder is imposed directly on the boundary-crossing measure and therefore does not contain any term such as $\sup_x |H_{c,b}^{(4)}(x)|$. $\square$

Lemma 6 (Localized argmax transfer). *Let $Q_n(u)$ be random upper-semicontinuous criteria on compact sets of $\mathbb{R}$. Suppose*

$$Q_n(u) \Rightarrow Q(u) = \mathbb{G}_\partial(u) - \frac{1}{2}\kappa u^2 + u\xi$$

in $\ell^\infty(K)$ for every compact K , where $\kappa > 0$, $\mathbb{G}_\partial$ is tight with continuous sample paths on compact sets, and Q has an almost surely unique maximizer. Suppose also that the sequence of near maximizers $\widehat{u}_n$ is tight and satisfies

$$Q_n(\widehat{u}_n) \geq \sup_u Q_n(u) - o_p(1).$$

Then

$$\widehat{u}_n \Rightarrow \arg \max_{u \in \mathbb{R}} Q(u).$$

Proof. Tightness allows restriction to a compact set with probability arbitrarily close to one. On that compact set, uniform convergence of Q_n to Q and uniqueness of the maximizer imply convergence of the argmax by the standard argmax continuous mapping theorem. The near-maximization error is $o_p(1)$, so it does not alter the limit. Letting the compact set expand completes the proof. $\square$

Lemma 7 (Boundary-kernel replacement orders). *Let $\widehat{X} - X = O_p(m^{-1/2})$ in local L^2 over every boundary tube $\{|X - c_0| \leq Ch\}$, uniformly over fixed C , and impose the derivative-envelope condition in Assumption 6(ii). Let K have bounded derivatives through order two. Then replacing X by $\widehat{X}$ in the boundary-kernel trace estimator contributes order $m^{-1/2}/h$, while replacing X by $\widehat{X}$ in the derivative trace-slope estimator contributes order $m^{-1/2}/h^2$.*

Proof. For the trace estimator,

$$K\left(\frac{\widehat{X} - c}{h}\right) - K\left(\frac{X - c}{h}\right) = K'(\xi) \frac{\widehat{X} - X}{h}$$

for an intermediate point ξ , so boundedness of K' gives an $O_p(m^{-1/2}/h)$ perturbation after averaging with bounded moments. For the derivative estimator, the kernel term is $h^{-2}K'((X - c)/h)$. Replacing X by $\widehat{X}$ changes K' by $K''(\xi)(\widehat{X} - X)/h$ and therefore contributes $m^{-1/2}/h^2$ at the level of the derivative estimator's stochastic order. This proves the stated replacement orders. $\square$

B Proofs of the main results

Proof of Theorem 1. If $\widehat{X} \geq c + r$, then $X \geq \widehat{X} - r \geq c$, so $\mathbf{1}\{X \geq c\} = 1$ for every latent state consistent with the calibration restriction. If $\widehat{X} < c - r$, then $X \leq \widehat{X} + r < c$, so $\mathbf{1}\{X \geq c\} = 0$ for every consistent latent state. If $c - r \leq \widehat{X} < c + r$, both values of $\mathbf{1}\{X \geq c\}$ are feasible: choose $X = c$ or slightly above to make it one, and choose $X < c$ to make it zero, respecting the radius restriction after arbitrarily small adjustments at boundary ties. A measurable selection is obtained by using the observed variables and an arbitrarily small independent tie-breaking perturbation; letting the perturbation vanish leaves the bounds unchanged.

Therefore, outside the boundary band the contribution is fixed and inside the band the researcher can only bound Γb over $b \in \{0, 1\}$. The minimum is $\min\{\Gamma, 0\} = \Gamma_-$ and the maximum is $\max\{\Gamma, 0\} = \Gamma_+$. Taking expectations gives the stated bounds.

Sharpness follows by measurable selection. On the upper-bound construction, set $X \geq c$ in the boundary band whenever $\Gamma > 0$ and set $X < c$ whenever $\Gamma \leq 0$. On the lower-bound construction, reverse the choices. These selections preserve the observed law of $(\widehat{X}, \Gamma)$ and satisfy $|X - \widehat{X}| \leq r$. Hence no narrower identified set is possible. The width formula follows from $\Gamma_+ - \Gamma_- = |\Gamma|$. $\square$

Proof of Lemma 1. Let $A \subset (0, \bar{a})$ and write $R_A = \int_A t^p f_X(-t) dt$. Since f_X is bounded above and below near the boundary, $R_A \asymp \int_A t^p dt$. Also $w(t) \leq \bar{K}t^\alpha$. The problem of maximizing $\int_A t^\alpha dt$ subject to $\int_A t^p dt \leq CR$ is a one-dimensional “bathtub” problem. Because $0 \leq \alpha < p$, the benefit-cost ratio $t^\alpha/t^p = t^{\alpha-p}$ is decreasing in t . Therefore any set that contains a point t_2 but omits a smaller point $t_1 < t_2$ can be improved weakly by replacing a small interval around t_2 with one around t_1 of the same t^p -cost. Iterating this exchange yields the initial interval $[0, a]$ as a maximizer up to null sets.

For $A = [0, a]$,

$$\int_0^a t^p dt = \frac{a^{p+1}}{p+1}, \quad \int_0^a t^\alpha dt = \frac{a^{\alpha+1}}{\alpha+1}.$$

The cost constraint gives $a \lesssim R^{1/(p+1)}$, and hence

$$\int_A w(t) dt \leq C \int_0^a t^\alpha dt \leq CR^{(\alpha+1)/(p+1)}.$$

The same interval attains the rate by the lower bounds on f_X and w , proving the lemma. $\square$

Proof of Theorem 2. The lower bound is constructive. Choose $a > 0$ and set

$$u_a(x) = -x \mathbf{1}\{-a \leq x < 0\}.$$

This perturbation flips exactly the interval $[-a, 0)$ into the treated region. Its L^p cost is

$$\mathbb{E}|u_a(X)|^p = \int_0^a t^p f_X(-t) dt \asymp a^{p+1}.$$

Set $a \asymp r^{p/(p+1)}$ so that $\mathbb{E}|u_a(X)|^p \leq r^p$. The welfare gain is

$$\int_0^a w(t) dt \asymp \int_0^a t^\alpha dt \asymp a^{\alpha+1} \asymp r^{p(\alpha+1)/(p+1)}.$$

This proves $\omega_p(r) \gtrsim r^{p(\alpha+1)/(p+1)}$.

For the upper bound, any point $X = -t < 0$ can be flipped upward only if $u(X) \geq t$. Let

$$A_u = \{t \in (0, \bar{a}) : u(-t) \geq t\}$$

be the set of flipped boundary distances. Since $u(-t)^p \geq t^p$ on A_u ,

$$\int_{A_u} t^p f_X(-t) dt \leq \int_{A_u} u(-t)^p f_X(-t) dt \leq r^p.$$

The local welfare gain is $\int_{A_u} w(t) dt$. Lemma 1 bounds it by

$$C(r^p)^{(\alpha+1)/(p+1)} = Cr^{p(\alpha+1)/(p+1)}.$$

Assumption 2 rules out a larger contribution away from the boundary for small budgets. This proves the matching upper bound. $\square$

Proof of Theorem 3. For a smooth direction h , write

$$\Theta_c(g_0 + th) - \Theta_c(g_0) = \mathbb{E}[\Gamma\{\mathbf{1}(X + th(Z) \geq c) - \mathbf{1}(X \geq c)\}].$$

Applying the first-order boundary expansion with $\Delta = th(Z)$ gives

$$\Theta_c(g_0 + th) - \Theta_c(g_0) = t f_X(c) \mathbb{E}[\Gamma h(Z) \mid X = c] + o(t).$$

This proves the directional derivative.

To show lack of L^2 continuity, choose smooth bump functions $h_j(Z) = \varphi(j(X - c))$ with $\varphi(0) = 1$ and compact support. Then $\mathbb{E}[h_j(Z)^2] \asymp j^{-1} \rightarrow 0$, so $h_j \rightarrow 0$ in $L^2(P_Z)$. However,

$$\mathbb{E}[\Gamma h_j(Z) \mid X = c] = \mathbb{E}[\Gamma \mid X = c] \varphi(0) = \mathbb{E}[\Gamma \mid X = c].$$

Thus $D\Theta_c(g_0)[h_j]$ does not converge to zero when $f_X(c) \mathbb{E}[\Gamma \mid X = c] \neq 0$. A continuous linear functional on $L^2(P_Z)$ must send every L^2 -null sequence to zero. Hence the derivative is not L^2 continuous and cannot have an L^2 Riesz representer. $\square$

Proof of Proposition 2. Let $h_j(Z) = \varphi(j(X - c_0))$ with φ smooth, compactly supported, and $\varphi(0) = 1$. Then $\|h_j\|_{L^2(P_Z)}^2 \asymp j^{-1} \rightarrow 0$. Because h_j is a function of X , the trace-slope

functional satisfies

$$\begin{aligned}\mathcal{S}_{c_0}(h_j) &= \partial_c \{f_X(c)m(c)\varphi(j(c - c_0))\}_{c=c_0} \\ &= \partial_c \{f_X(c)m(c)\}_{c=c_0} \varphi(0) + f_X(c_0)m(c_0)j\varphi'(0).\end{aligned}$$

The second term is zero because $m(c_0) = 0$, while the first term is nonzero by assumption. Thus $h_j \rightarrow 0$ in $L^2(P_Z)$ but $\mathcal{S}_{c_0}(h_j)$ does not converge to zero. Hence the trace-slope cannot be continuously extended as a global $L^2(P_Z)$ functional. $\square$

Proof of Theorem 4. Apply Lemma 5 with the signed conditional boundary-crossing measure

$$\mu_x(dd) = f_X(x)E[\Gamma \mathbf{1}\{\Delta \in dd\} \mid X = x].$$

The lemma gives directly

$$B_g(c) = h_1(c) - \frac{1}{2}h_2'(c) + \frac{1}{6}h_3''(c) + O(r^4).$$

This proof uses the exact oriented interval crossed by the discontinuous rule. It therefore does not require bounding derivatives of a smoothed indicator. The fourth-order condition in Assumption 4 is a primitive condition on the signed boundary-crossing measure, equivalently on the boundary moments and their local Taylor remainder, rather than a bound involving $H_{c,b}^{(4)}$. The second-order expression follows by absorbing the third and higher terms into $o(r^2)$ under the stated small-perturbation condition. The negative sign on the second-order term is a consequence of the oriented interval identity: a positive perturbation brings mass from just below the cutoff, while the derivative term differentiates the second boundary moment with respect to the cutoff. $\square$

Proof of Theorem 5. Let $d = c_{g,q} - c_q$. The treatment-share restriction is

$$0 = P(X + \Delta \geq c_q + d) - P(X \geq c_q).$$

Equivalently, with the perturbation $\Delta - d$, this is

$$0 = \mathbb{E}[\mathbf{1}\{X + \Delta - d \geq c_q\} - \mathbf{1}\{X \geq c_q\}].$$

Apply Theorem 4 with $\Gamma = 1$ and perturbation $\Delta - d$. The first-order term is

$$f_X(c_q)\mathbb{E}[\Delta - d \mid X = c_q] = f_X(c_q)\{\mathbb{E}[\Delta \mid X = c_q] - d\}.$$

The second-order and remainder terms are $O(r^2 + d^2)$ uniformly in a neighborhood of zero. Since $f_X(c_q) > 0$ and $d = O(r)$ by monotonicity of the distribution around c_q , the equation implies

$$d = \mathbb{E}[\Delta \mid X = c_q] + O(r^2).$$

Now write the welfare distortion of the budget-constrained generated rule as

$$\mathbb{E}[\Gamma\{\mathbf{1}(X + \Delta \geq c_q + d) - \mathbf{1}(X \geq c_q)\}] = \mathbb{E}[\Gamma\{\mathbf{1}(X + \Delta - d \geq c_q) - \mathbf{1}(X \geq c_q)\}].$$

Apply the first-order part of Theorem 4 with perturbation $\Delta - d$:

$$f_X(c_q)\mathbb{E}[\Gamma(\Delta - d) \mid X = c_q] + O(r^2).$$

Substituting $d = \mathbb{E}[\Delta \mid X = c_q] + O(r^2)$ gives

$$\begin{aligned} f_X(c_q)\{\mathbb{E}[\Gamma\Delta \mid X = c_q] - \mathbb{E}[\Gamma \mid X = c_q]\mathbb{E}[\Delta \mid X = c_q]\} + O(r^2) \\ = f_X(c_q)\text{Cov}(\Gamma, \Delta \mid X = c_q) + O(r^2), \end{aligned}$$

which proves the proposition. $\square$

Proof of Theorem 6. Let $\delta_m = \widehat{g}_m - g_0$ and write the common-shift component as $a(\delta_m) = E[\delta_m(Z) \mid X = c_0]$. The local coordinate u is the latent-equivalent cutoff coordinate, so the corresponding raw generated-score cutoff is

$$c_{\text{raw}}(u) = c_0 + a(\delta_m) + n^{-1/3}u.$$

Consider the localized objective

$$Q_{n,m}(u) = n^{2/3}\{\widehat{M}_{n,m}(c_0 + a(\delta_m) + n^{-1/3}u) - \widehat{M}_{n,m}(c_0 + a(\delta_m))\}.$$

Then

$$\arg \max_u Q_{n,m}(u) = n^{1/3}(\widehat{c}_{n,m} - a(\delta_m) - c_0) = n^{1/3}(\widetilde{c}_{n,m} - c_0).$$

Decompose it into four terms:

$$Q_{n,m}(u) = \widehat{A}_{n,m}(u) + B_n(u) + C_{n,m}(u) + R_{n,m}(u),$$

where $\widehat{A}_{n,m}$ is the centered local empirical process written in the residualized generated coordinate, B_n is the deterministic population curvature, $C_{n,m}$ is the residualized generated-state mean drift, and $R_{n,m}$ collects nuisance and higher-order terms. This decomposition is important: at $m \asymp n^{2/3}$, the centered difference between an oracle-coordinate process and a generated-coordinate process need not be negligible. The proof therefore uses the generated-coordinate local process in Assumption 5(i), rather than replacing it by an oracle-coordinate process.

By Assumption 5, $\widehat{A}_{n,m}(u) \Rightarrow \mathbb{G}_\partial(u)$ in ℓ^∞ on compact sets, conditionally on the validation sample and asymptotically independently of the validation influence function. The curvature condition gives

$$B_n(u) = n^{2/3}\{M(c_0 + n^{-1/3}u) - M(c_0)\} \rightarrow -\frac{1}{2}\kappa u^2.$$

For the generated-state drift, use the residualized distortion in Assumption 5(iii). The common-shift component has already been absorbed by the raw cutoff coordinate $c_0 + a(\delta_m) + n^{-1/3}u$, so the remaining population distortion is

$$C_{n,m}(u) = n^{2/3}\{B_{v_m}(c_0 + n^{-1/3}u) - B_{v_m}(c_0)\} = n^{1/3}u\mathcal{T}_{c_0}^\perp(\widehat{g}_m - g_0) + o_p(1).$$

Using the first-stage expansion,

$$\mathcal{T}_{c_0}^\perp(\widehat{g}_m - g_0) = \eta_m \mathcal{T}_{c_0}^\perp(b) + \frac{1}{m} \sum_{j=1}^m \chi^\perp(S_j) + o_p(n^{-1/3}).$$

The deterministic part satisfies $n^{1/3}\eta_m \mathcal{T}_{c_0}^\perp(b) \rightarrow \beta_b$. The stochastic part satisfies

$$n^{1/3} \frac{1}{m} \sum_{j=1}^m \chi^\perp(S_j) = \sqrt{\frac{n^{2/3}}{m}} \left(\frac{1}{\sqrt{m}} \sum_{j=1}^m \chi^\perp(S_j) \right) \Rightarrow \sqrt{\lambda_g} \sigma_{T,\perp} N_g.$$

Independence of the evaluation and validation samples gives independence of N_g and $\mathbb{G}_\partial$. The remainder is negligible by assumption. Therefore $Q_{n,m}$ converges to the stated limiting criterion on every compact set. Assumption 5 imposes tightness of the localized maximizers

and uniqueness of the limiting maximizer. Lemma 6 then transfers compact convergence of the localized criteria into convergence of the argmax. This proves the result. $\square$

Proof of Theorem 8. The proof has four steps.

Step 1: calibration expansion. By the Z-estimator conditions in Assumption 6,

$$0 = \frac{1}{m} \sum_{j=1}^m \psi(S_j; \widehat{\theta}) = \frac{1}{m} \sum_{j=1}^m \psi(S_j; \theta_0) + H(\widehat{\theta} - \theta_0) + o_p(m^{-1/2}).$$

Solving gives

$$\widehat{\theta} - \theta_0 = -H^{-1} \frac{1}{m} \sum_{j=1}^m \psi(S_j; \theta_0) + o_p(m^{-1/2}).$$

Step 2: fixed-policy trace. A Taylor expansion of $q_{\widehat{\theta}}(Z)$ around θ_0 gives

$$q_{\widehat{\theta}}(Z) - q_{\theta_0}(Z) = \dot{q}_{\theta_0}(Z)^\top (\widehat{\theta} - \theta_0) + R_q(Z),$$

where $\mathbb{E}|R_q(Z)| = O_p(\|\widehat{\theta} - \theta_0\|^2) = o_p(m^{-1/2})$. Applying the first-order boundary expansion to

$$\Delta(Z) = q_{\widehat{\theta}}(Z) - q_{\theta_0}(Z)$$

yields

$$B_m(c) = f_X(c) \mathbb{E}[\Gamma \Delta(Z) \mid X = c] + o_p(m^{-1/2}).$$

Substituting the Taylor expansion of q gives

$$B_m(c) = A(c)^\top (\widehat{\theta} - \theta_0) + o_p(m^{-1/2}).$$

Using Step 1,

$$B_m(c) = -A(c)^\top H^{-1} \frac{1}{m} \sum_{j=1}^m \psi(S_j; \theta_0) + o_p(m^{-1/2}) = \frac{1}{m} \sum_{j=1}^m b_c(S_j) + o_p(m^{-1/2}).$$

The central limit theorem gives the stated normal limit and variance.

Step 3: learned-cutoff trace-slope. The raw trace-slope operator differentiates the fixed trace with respect to the cutoff:

$$\mathcal{T}_{c_0}(q_{\widehat{\theta}} - q_{\theta_0}) = \partial_c \{f_X(c) \mathbb{E}[\Gamma \Delta(Z) \mid X = c]\}_{c=c_0} + o_p(m^{-1/2}).$$

Using the same Taylor expansion of q gives

$$\mathcal{T}_{c_0}(q_{\widehat{\theta}} - q_{\theta_0}) = \dot{A}(c_0)^\top (\widehat{\theta} - \theta_0) + o_p(m^{-1/2}).$$

The common-shift component equals

$$a(q_{\widehat{\theta}} - q_{\theta_0}) = D(c_0)^\top (\widehat{\theta} - \theta_0) + o_p(m^{-1/2}).$$

Hence the policy-relevant trace-slope is

$$\mathcal{T}_{c_0}^\perp(q_{\widehat{\theta}} - q_{\theta_0}) = \dot{A}_\perp(c_0)^\top (\widehat{\theta} - \theta_0) + o_p(m^{-1/2}).$$

Substituting Step 1 gives

$$\mathcal{T}_{c_0}^\perp(\widehat{g} - g_0) = \frac{1}{m} \sum_{j=1}^m \chi_{c_0}^\perp(S_j) + o_p(m^{-1/2}),$$

with $\chi_{c_0}^\perp(S) = -\dot{A}_\perp(c_0)^\top H^{-1} \psi(S; \theta_0)$. The stated variance and CLT follow.

Step 4: connection to the learned-cutoff local experiment. Theorem 6 contains the local term

$$n^{1/3} \mathcal{T}_{c_0}^\perp(\widehat{g} - g_0).$$

By Step 3,

$$n^{1/3} \mathcal{T}_{c_0}^\perp(\widehat{g} - g_0) = \sqrt{\frac{n^{2/3}}{m}} \left(\frac{1}{\sqrt{m}} \sum_{j=1}^m \chi_{c_0}^\perp(S_j) \right) + o_p(1).$$

If $n^{2/3}/m \rightarrow \lambda_g$, this converges to $\sqrt{\lambda_g} \sigma_{T,\perp} N_g$, completing the proof. $\square$

Proof of Theorem 9. Uniformly over the boundary-local order, the calibrated expansion gives

$$\delta_{\widehat{\theta}}(Z) = q_{\widehat{\theta}}(Z) - q_{\theta_0}(Z) = \dot{q}_{\theta_0}(Z)^\top (\widehat{\theta} - \theta_0) + o_p(m^{-1/2}), \quad \nu_{\widehat{\theta}} = \delta_{\widehat{\theta}} - a(\delta_{\widehat{\theta}}).$$

The signed boundary-crossing expansion applied to $v_{\hat{\theta}}$ gives, uniformly for compact u -sets,

$$\begin{aligned} B_{v_{\hat{\theta}}}(c_0 + n^{-1/3}u) - B_{v_{\hat{\theta}}}(c_0) \\ = n^{-1/3}u \partial_c \{f_X(c)E[\Gamma v_{\hat{\theta}}(Z) \mid X = c]\}_{c=c_0} \\ + o_p(n^{-1/3}m^{-1/2}) + o_p(n^{-2/3}). \end{aligned}$$

Because $a(\delta_{\hat{\theta}}) = D(c_0)^\top(\hat{\theta} - \theta_0) + o_p(m^{-1/2})$, the derivative equals

$$\{\dot{A}(c_0) - \kappa D(c_0)\}^\top(\hat{\theta} - \theta_0) + o_p(m^{-1/2}) = \dot{A}_\perp(c_0)^\top(\hat{\theta} - \theta_0) + o_p(m^{-1/2}).$$

Multiplying by $n^{2/3}$ proves the residualized drift display. Finally, the Z-estimator expansion $\hat{\theta} - \theta_0 = -H^{-1}m^{-1} \sum_{j=1}^m \psi(S_j; \theta_0) + o_p(m^{-1/2})$ implies

$$\sqrt{m} \mathcal{T}_{c_0}^\perp(q_{\hat{\theta}} - q_{\theta_0}) \Rightarrow N\{0, \dot{A}_\perp(c_0)^\top H^{-1} \Omega H^{-\top} \dot{A}_\perp(c_0)\}.$$

Together with Assumption 5(i),(ii),(iv), this verifies the calibrated learned-cutoff local experiment. $\square$

Proof of Theorem 7. Work inside the product local class $\mathcal{P}_{n,m}(C)$ from Theorem 7. Consider the validation-only restricted experiment in which the evaluation sample is generated from a fixed oracle law and only the validation calibration parameter varies. Take local alternatives $\theta_\pm = \theta_0 \pm h/\sqrt{m}$. LAN implies that the validation experiments under θ_+ and θ_- have bounded total variation distance for fixed h ; because the evaluation law is identical across the two alternatives, the same bound holds in the full product experiment. Hence there is a constant $\pi_0 > 0$ such that every estimator confuses the two alternatives with probability at least π_0 along a subsequence. The usual two-point inequality then gives a lower bound equal to a fixed fraction of the separation between the two implied policy-equivalent cutoffs.

The oracle cutoff changes with θ . By the implicit function theorem applied to the first-order condition for $M_\theta(c)$, and using the nonzero non-reparameterized trace-slope $\tau^\perp$,

$$c(\theta_0 + t) = c_0 + \frac{\tau^\perp}{\kappa}t + o(t).$$

Thus $|c(\theta_+) - c(\theta_-)| \asymp m^{-1/2}$. Le Cam's two-point method implies that the minimax expected cutoff error is at least a constant multiple of $m^{-1/2}$. Since the objective has local quadratic curvature, cutoff error e induces regret at least of order e^2 locally, giving a

validation-stage regret lower bound of order m^{-1} .

For the evaluation component, use the complementary restricted experiment in which the validation channel is fixed at the true state and only the evaluation distribution varies locally. This is a subexperiment of the full model, so any estimator valid in the full model must also work there. The ordinary threshold-learning lower bound gives regret of order $n^{-2/3}$ even when the state is known; this is the minimax lower-bound counterpart to the Kim–Pollard local experiment and can be obtained from the standard two-point margin construction used in plug-in classification lower bounds.

For the validation component, restrict the evaluation distribution and welfare curvature to a fixed member of the class and vary only the validation-calibration submodel above. Because the evaluation and validation samples are independent by construction, both restricted experiments are legitimate subexperiments of the full experiment. Any estimator in the full experiment induces an estimator in each restricted experiment. Hence its worst-case risk over the full class is at least the maximum of the two restricted minimax risks,

$$C \max\{n^{-2/3}, m^{-1}\}.$$

This is the formal finite-sample minimax lower-bound statement. It is stated as a max lower bound. The local asymptotic product experiment is more informative about constants: take an independent prior over the evaluation threshold-local parameter and the validation-local calibration parameter. Conditional on the two independent observations, posterior variances add for the sum of the oracle cutoff component and the trace-slope component. Hence the Bayes risk in the product limit experiment is the sum of the two component Bayes risks. This explains the additive variance in Theorem 6. The max lower bound above is the conservative statement needed for the validation frontier, because the first-stage component is negligible relative to the evaluation-sample cube-root component if and only if $m^{-1} = o(n^{-2/3})$, equivalently $m \gg n^{2/3}$. $\square$

References

- ALATAS, V., A. BANERJEE, R. HANNA, B. A. OLKEN, AND J. TOBIAS (2012): “Targeting the Poor: Evidence from a Field Experiment in Indonesia,” *American Economic Review*, 102, 1206–1240.
- ANDREWS, D. W. K. (1992): “Generic Uniform Convergence,” *Econometric Theory*, 8, 241–257.

- ARGAÑARAZ, F. AND J. C. ESCANCIANO (2025): “Debiased Machine Learning for Unobserved Heterogeneity: High-Dimensional Panels and Measurement Error Models,” *arXiv preprint arXiv:2507.13788*.
- ASH, E. AND S. HANSEN (2023): “Text Algorithms in Economics,” *Annual Review of Economics*, 15, 659–688.
- ATHEY, S. AND S. WAGER (2021): “Policy Learning with Observational Data,” *Econometrica*, 89, 133–161.
- AUDIBERT, J.-Y. AND A. B. TSYBAKOV (2007): “Fast Learning Rates for Plug-in Classifiers,” *The Annals of Statistics*, 35, 608–633.
- BARTALOTTI, O., Q. BRUMMET, AND S. DIETERLE (2021): “A Correction for Regression Discontinuity Designs with Group-Specific Mismeasurement of the Running Variable,” *Journal of Business & Economic Statistics*, 39, 833–848.
- BATTAGLIA, L., T. CHRISTENSEN, S. HANSEN, AND S. SACHER (2024): “Inference for Regression with Variables Generated from Unstructured Data,” *CESifo Working Paper*.
- CATTANEO, M. D., M. JANSSON, AND K. NAGASAWA (2020): “Bootstrap-Based Inference for Cube Root Asymptotics,” *Econometrica*, 88, 2203–2219.
- CHERNOZHUKOV, V., D. CHETVERIKOV, M. DEMIRER, E. DUFLO, C. HANSEN, W. NEWEY, AND J. ROBINS (2018): “Double/Debiased Machine Learning for Treatment and Structural Parameters,” *The Econometrics Journal*, 21, C1–C68.
- CRIPPA, F. (2025): “Regret Analysis in Threshold Policy Design,” *Journal of Econometrics*, 249, 105998.
- DAVEZIES, L. AND T. LE BARBANCHON (2017): “Regression Discontinuity Design with Continuous Measurement Error in the Running Variable,” *Journal of Econometrics*, 200, 260–281.
- FANG, Y., G. RIDDER, AND H. XIE (2025): “Semiparametric Efficiency in Policy Learning with General Treatments,” *arXiv preprint arXiv:2512.19230*.
- GENTZKOW, M., B. KELLY, AND M. TADDY (2019): “Text as Data,” *Journal of Economic Literature*, 57, 535–574.
- HAHN, J. AND G. RIDDER (2013): “Asymptotic Variance of Semiparametric Estimators with Generated Regressors,” *Econometrica*, 81, 315–340.
- HIRANO, K. AND J. R. PORTER (2009): “Asymptotics for Statistical Treatment Rules,” *Econometrica*, 77, 1683–1701.

- KIM, J. AND D. POLLARD (1990): “Cube Root Asymptotics,” *The Annals of Statistics*, 18, 191–219.
- KITAGAWA, T. AND A. TETENOV (2018): “Who Should Be Treated? Empirical Welfare Maximization Methods for Treatment Choice,” *Econometrica*, 86, 591–616.
- LUDWIG, J., S. MULLAINATHAN, AND A. RAMBACHAN (2026): “Large Language Models: An Applied Econometric Framework,” *Annual Review of Economics*, 18.
- MAMMEN, E. AND A. B. TSYBAKOV (1999): “Smooth Discrimination Analysis,” *The Annals of Statistics*, 27, 1808–1829.
- MANSKI, C. F. (2004): “Statistical Treatment Rules for Heterogeneous Populations,” *Econometrica*, 72, 1221–1246.
- MBAKOP, E. AND M. TABORD-MEEHAN (2021): “Model Selection for Treatment Choice: Penalized Welfare Maximization,” *Econometrica*, 89, 825–848.
- MURPHY, K. M. AND R. H. TOPEL (2002): “Estimation and Inference in Two-Step Econometric Models,” *Journal of Business & Economic Statistics*, 20, 88–97.
- NEWKEY, W. K. (1994): “The Asymptotic Variance of Semiparametric Estimators,” *Econometrica*, 62, 1349–1382.
- NEWKEY, W. K. AND D. MCFADDEN (1994): “Large Sample Estimation and Hypothesis Testing,” *Handbook of Econometrics*, 4, 2111–2245.
- OPOCHER, G. (2026): “Better Measurement or Larger Samples? Data Collection for Policy Learning with Unobserved Heterogeneity,” *arXiv preprint arXiv:2604.07181*.
- PAGAN, A. (1984): “Econometric Issues in the Analysis of Regressions with Generated Regressors,” *International Economic Review*, 25, 221–247.
- PEI, Z. AND Y. SHEN (2017): *The Devil Is in the Tails: Regression Discontinuity Design with Measurement Error in the Assignment Variable*, vol. 38 of *Advances in Econometrics*, Emerald Publishing Limited.
- SUN, R. (2026): “Bootstrap-Based Inference for Cube Root Asymptotics with High-Frequency-Generated Regressors,” *Available at SSRN 6719545*.
- TSYBAKOV, A. B. (2004): “Optimal Aggregation of Classifiers in Statistical Learning,” *The Annals of Statistics*, 32, 135–166.